MIL-D-52091C(ME)
26 June 1970
SUPERSEDING
MIL-D-52091B(MO)
1 April 1966

MILITARY SPECIFICATION

DITCHING MACHINE: DIESEL-ENGINE-DRIVEN; WHEEL-MOUNTED; LADDER

1. SCOPE

1.1 This specification covers one type of diesel-engine-driven, self-propelled, vertical-boom, rubber-tired, ladder type, ditching machine.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

O-A-548	- Antifreeze, Ethylene Glycol, Inhibited.
JJ-B-185	- Belt, Seat, Passenger Type, Automotive.
QQ-S-781	- Strapping, Steel, Flat and Seals.
UU-T-81	- Tags, Shipping and Stock.
VV-F-800	- Fuel Oil, Diesel.
ZZ-H-428	- Hose, Rubber: Preformed and Straight; Hose, Preformed: Flexible Wire-Reinforced (for the Coolant Systems of Automotive and Other Liquid Cooled Engines).
ZZ-T-1083	- Tires, Pneumatic, Low Speed, Off Highway and Industrial.
KKK-L-370	- Lining, Friction, (Clutch and Brake; Metallic, Metal-Ceramic and Semimetallic).
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-T-60	- Tape: Pressure-Sensitive Adhesive, Waterproof, for Packaging.

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Military

MIL-C-104	- Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	- Preservation, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant, (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-I-631	- Insulation, Electrical, Synthetic-Resin Composition, Nonrigid.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-1283	- Can, Gasoline, Military 5-Gallon.
MIL-L-2104	- Lubricating Oil, Internal-Combustion Engine, Heavy Duty.
MIL-L-2105	- Lubricating Oil, Gear, Multipurpose.
MIL-L-3661	- Lampholders; Lights, Indicator; Indicator-Light Housings; and Lenses, Indicator Light, General Specification for.
MIL-L-3661/7	- Lights, Indicator (Housings) (Type LH75).
MIL-L-3661/15	- Lenses, Indicator Light (Style LC15).
MIL-G-3787	- Glass, Laminated, Flat; (Except Aircraft).
MIL-G-3859	- Grease Guns, Hand, High-Pressure, Lever-Operated; Cartridge and Bulk Loading (14 Ounce and 21 Ounce Capacity).
MIL-L-6085	- Lubricating Oil: Aircraft Instrument, Low Volatility.
MIL-S-8660	- Silicone Compound.
MIL-L-10295	- Lubricating Oil, Internal-Combustion Engine, Sub-Zero.
MIL-L-10324	- Lubricating Oil, Gear, Sub-Zero.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-E-11276	- Engines, Diesel: Industrial, Medium- and High-Speed, General Specification for.
MIL-L-13071	- Lamp Housing, Resilient Mounting, Construction Machinery.
MIL-C-13489	- Cover and End Curtains; Cargo Body (for Military Vehicles).
MIL-I-15126	- Insulation Tape, Electrical, Pressure Sensitive Adhesive and Pressure Sensitive Thermosetting Adhesive.

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MIL-H-15424 - Hand Tools; Packaging of.
MIL-C-21567 - Compound, Silicone, Soft Film.
MIL-G-23827 - Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-T-45043 - Towbars, Motor Vehicle, V-Type.
MIL-T-46786 - Tanks, Fuel, Engine, General Requirements for.
MIL-A-52363 - Air Cleaners, Intake: Dry-Type (for Internal Combustion Engines).
MIL-M-52590 - Mufflers, Exhaust, Internal Combustion Engine, General Purpose.
MIL-R-52602 - Reservoir Assembly: Oil, Hydraulic, Sealed.
MIL-P-52675 - Pumps, Hydraulic, Oil, Constant Displacement, 2000 PSI.

STANDARDS

Federal

FED. TEST METHOD STD. No. 191 - Textile Test Methods.

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U. S. Military Property.
MIL-STD-281 - Automobiles, Trucks, Truck-Tractors, Trailers and Trailer Dollies; Preservation and Packaging of.
MIL-STD-421 - Chain, Roller; Power Transmission and Conveyor, Flat Link Plates, Single Pitch, Single and Multiple Strand; Connecting Links and Attachment Links.
MIL-STD-642 - Identification Marking of Administrative, Combat, and Tactical Transport Vehicles.
MIL-STD-646 - Electrical Circuit Number Identification for Military Vehicles.
MIL-STD-831 - Test Reports, Preparation of.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.
MIL-STD-1224 - Visual Inspection Guide for Pneumatic Tires (Non-Aircraft).
MIL-STD-1400 - Engines, Gasoline or Diesel, Methods of Test.

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MS15001	- Fittings, Lubrication, (Hydraulic) Surface Check, 1/4 - 28 Taper Threads, Steel, Type I.
MS15002	- Fittings, Lubrication, (Hydraulic) Surface Check, Straight Threads, Steel, Type II.
MS15003	- Fittings, Lubrication (Hydraulic) Surface Check, 1/8 Pipe Threads, Steel, Type III.
MS24207	- Coupling, Grease Gun.
MS24544	- Indicator, Liquid Quantity - 24 Volt D. C. Engine Fuel.
MS27142	- Connector, Plug, Electrical - Pin Contact No. 12, 14 & 16 A.W.G., Waterproof.
MS27143	- Connector, Plug, Electrical - Socket Contact, No. 12 A.W.G., Waterproof.
MS27144	- Connector, Plug, Electrical - Socket Contact, No. 14 & 16 A.W.G., Waterproof.
MS27145	- Connector, Plug, Electrical - No. 14 A.W.G., Double Contact, Waterproof.
MS27146	- Connector, Receptacle, Electrical - No. 12, 14 & 16 A.W.G., Double Contact, Waterproof.
MS27147	- Adapter, Connector - 3 Way, No. 12, 14 & 16 A.W.G. Waterproof.
MS27148	- Pin Contact, Electrical Connector - No. 12, 14 & 16 A.W.G.
MS27152	- Switch, Pressure-Warning, Low Air Pressure, 6 PSI, SPST, Waterproof.
MS33800	- Connector, Receptacle, Electrical-Pin Contact, No. 12, 14 & 16 A.W.G., Waterproof.
MS35000	- Battery, Storage, Lead-Acid, Waterproof.
MS35423	- Light, Marker, Clearance - Service.
MS35424	- Light, Marker, Clearance - Blackout, 24 Volt.
MS35647	- Padlocks, Key Operated.
MS35746	- Coupling, Automotive - Air Brake Lines.
MS35802	- Filter Elements, Fluid, Pressure-Oil, Full-Flow.
MS35829	- Hinge, Continuous (Piano), .075 Thickness, .187 and .250 Pin Dia.
MS35844	- Plug, Machine Thread; Magnetic (Drain).
MS39020	- Band, Marker, Blank.
MS39060	- Switch, Rotary, 24-Volt D. C., Waterproof.
MS39061	- Switch, Toggle - SPST, SPDT, 24 Volt DC, 25 Amp (Waterproof).
MS39062	- Circuit Breakers: Thermal, 30 Volt DC, Automatic, Waterproof, 15 to 30 Amp.
MS39248	- Regulator, Alternator, 28-Volt, 35 Ampere.

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MS39249 - Alternator, 28-Volt, 35 Ampere.
MS39250 - Regulator, Alternator, 28-Volt, 35 Amperes, For Use With Self-Load Limiting Alternators.
MS39251 - Alternator, 28-Volt, 35-Ampere, Double Mounting Flange.
MS39252 - Regulator, Alternator, 28-Volt, 35 Ampere.
MS39253 - Alternator, 28-Volt, 35-Ampere, Single Mounting Flange.
MS39281 - Belting, Rubber Covered; Conveyor, Ditching Machine Discharge with Cleats and Notched Guide Strip; Fully Molded.
MS39317 - Hose Assemblies; Hydraulic, Double Wire Braid Reinforcement, with Reusable Couplings.
MS39318 - Hose Assemblies; Hydraulic, 4-Spiral Wrap Reinforcement, with Reusable Couplings.
MS39321 - Fittings, Flared Tube, 37 Degree, Steel.
MS49006 - Plugs, Pipe, Magnetic-Headless; Aluminum, Iron or Steel.
MS51113 - Switch, Vehicular Lights: 24 Volt D. C.; (Waterproof).
MS51301 - Horn, Fluid Operated - Electrically Controlled, 24-Volt D. C. (Dual).
MS51303 - Light, Marker, Clearance - Blackout, 24 Volt.
MS51318 - Headlight: Blackout, 24 Volt, Waterproof.
MS51330 - Stop Light - Taillight, Vehicular - 24 Volt, Blackout Tail, Blackout Stop.
MS53046 - Retainer, Battery - Automotive, Tactical Vehicles.
MS53063 - Indicator, Air Cleaner, Intake; Restriction Mechanical Type.
MS53067 - Evaporator, Alcohol-Air Brake System.
MS75057 - Connector, Plug, Electrical-Slave, 2 Contact, 24 Volt, Waterproof.
MS75058 - Connector, Receptacle, Electrical-Slave, 2 Contact, 24 Volt, Waterproof.
MS500004 - Pad Eye, Vehicular Towing Lug.

DRAWINGS

ME

DL13201E3200 - Ditching Machine: Diesel-Engine-Driven; Wheel Mounted; Ladder, Bucket Line.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN WELDING SOCIETY

Standard Qualification Procedure.

(Application for copies should be addressed to the American Welding Society, 345 East 47th Street, New York, N. Y. 10017.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, N. Y. 10017.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street NW, Washington, D. C. 20036.)

DEPARTMENT OF TRANSPORTATION

Motor Carrier Safety Regulations.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

DEPARTMENT OF HEALTH, EDUCATION AND WELFARE (HEW)

Control of Air Pollution From New Motor Vehicles and New Motor Vehicle Engines.

(Applications for copies should be sent to Department of Health, Education, and Welfare, National Air Pollution Control Administration, Publications Unit, 5600 Fisher Lane, Rockville, Maryland. 20852.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS

B117 - Method of Salt Spray (Fog) Testing.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

STATE OF CALIFORNIA DEPARTMENT OF MOTOR VEHICLES

California Vehicle Code.

(Application for copies should be addressed to the State of California Department of Motor Vehicles, P. O. Box 1319, Sacramento, California 95806.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Tariff Publishing Officer, Room 202, Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

SOCIETY OF AUTOMOTIVE ENGINEERS

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Two Pennsylvania Plaza, New York, N. Y. 10001.)

TIRE AND RIM ASSOCIATION, INC.

Yearbook, with Military Supplement.

(Application for copies should be addressed to the Tire and Rim Association, Inc., Comand Building, 34 North Hawkins Avenue, Akron, Ohio 44313.)

U. S. WATERWAYS EXPERIMENTAL STATION

Technical Memorandum No. 3-357 - The Unified Soil Classification.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

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3. REQUIREMENTS

3.1 Description. The ditching machine shall be as follows: (1) self-propelled; (2) diesel-engine-driven; (3) front-wheel-steered; (4) supported by four pneumatic tires; and (5) 4-wheel drive (see 3.10). The ditching machine shall be equipped with a rear-mounted digging mechanism and a reversing, side-discharge, belt-type spoil conveyor.

3.1.1 Performance. The ditching machine (hereinafter called "ditcher") shall dig trenches 24 inches plus or minus 1/2 inch wide and from 0 to 6 feet deep. When operating on a flat, horizontal surface and digging in undisturbed CL-type or ML-type soil (WES, Technical Memorandum No. 3-357, Unified Soil Classification System), the ditcher shall: (1) dig at a depth of 4 feet at a rate of not less than 20 feet per minute; (2) dig with all walls, including end walls, vertical within 5 degrees when digging on level ground; and (3) dig curved trenches to a depth of 4 feet (follow-up scraper raised) with a radius not greater than 40 feet. The ditcher shall dig as specified herein on the following grades: (a) sidehill at 17.5 percent, (b) uphill at 21 percent, and (c) downhill at 27 percent.

3.1.2 Dimensions, weights, and capabilities. The ditcher shall conform to table I. The gross vehicle weight (gvw) shall include a full complement of fuel, lubricants, hydraulic oil, and coolants; special tools furnished with the ditcher; and not less than 350 pounds of personnel weight in the cab.

Table I. Dimensions, Weights, and Capabilities

Description	Travel	In Transport (Truck or Rail)
Dimensions		
Overall length, ft. (max.)	30	32
Overall width, in. (max.)	108	-
Overall height, in. (max.)	150	126
Wheelbase, in. (min.)	130	-
Ground clearance		
Under chassis, in. (min.)	13	-
Center of gravity		
Vertical, above ground level, in. (max.)	60	57
Longitudinal, percent of wheelbase, forward from center of rear axle, (min.)	36	-
Angle of approach, SAE J894, degrees (min.)	18	-
Angle of departure, SAE J894, with bucket teeth at the lowest position on boom, degrees (min.)	18	-

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Table I. Dimensions, Weights, and Capabilities (cont'd)

Description	Travel	In Transport (Truck or Rail)
Angle of interference, SAE J894, degrees (max.)	160	-
Weights		
Max. gvw, lb.	50,000	-
Min. gvw, lb.	34,000	-
Max. rear-axle weight when ditching, lb.	38,000	-
Performance		
Speeds*		
High gear forward, mph (min.)	27.0	-
Low gear forward , mph (min.)	4.3	-
Reverse gear, mph (min.)	3.5	-
Vehicle clearance circle, ft. dia. (max.)	80	-

* At an engine speed not exceeding the governed speed for the continuous horsepower rating at which the engine qualified.

3.2 First article.

3.2.1 Preproduction model. The supplier shall furnish a ditcher for examination and test within the time frame specified to prove, prior to starting production, that his production methods and choice of design detail will produce ditchers that comply with the requirements of this specification. Examination and tests shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.5). Evidence of nonconformance or failures occurring during the preproduction examination and testing shall be submitted to the contracting officer for a determination as to acceptability of the proposed corrective action and as to the retesting to be performed to establish conformance.

3.2.2 Hydraulic system components. Performance of hydraulic system components without malfunction or failure during the ditcher operating tests and performance tests shall constitute an acceptable test of the hydraulic components. In the event of hydraulic malfunction or failure, the failing component shall be tested in accordance with the applicable portion of Test Method No. 31 as specified in 4.6.2.35, or in accordance with the referenced specification, to support continued use of the make and model component selected by the ditcher supplier.

3.3 Initial production. Unless otherwise specified, the supplier shall furnish to the Government one or more ditchers for testing as specified in 4.4.

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3.4 Material. Material shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be those used for the same purpose on a ditcher of comparable size regularly sold on the commercial market and shall be subject to all provisions of this specification. Components, particularly of the electrical, fuel, and exhaust systems, shall resist harmful effects of dust, fresh or salt water, rain, and corrosion. To the maximum practicable extent, materials, and treatments shall be utilized to provide resistance to corrosion.

3.4.1 Threaded parts. Threaded parts shall be in accordance with Handbook H28.

3.4.2 Bearings. Antifriction bearings in the digging mechanism shall be mounted against shoulders.

3.5 Operational requirements.

3.5.1 Use. The ditcher shall withstand without damage or permanent deformation the usage encountered in military service such as:

(a) Crossing roadless, rough terrain including gullies causing the front axle to oscillate to the stops in both directions (see 3.10.5); fording 36-inch-deep streams; and negotiating uphill and downhill grades from 0 to 27 percent and sidehill slopes from 0 to 17.5 percent.
(b) Operating in hot, humid, tropical countries.
(c) Operating at altitudes ranging from sea level to 5,000 feet elevation.
(d) Driving at maximum travel speed in convoy and troop movement.
(e) Towing by another vehicle.
(f) Storing for prolonged periods unsheltered and exposed to weather conditions specified herein.
(g) Impact forces encountered in air and rail shipments.
(h) Excavating straight and curved trenches, foxholes, and utility ditches in CL- and ML-type soils, clay-gravel, shale, and ground frozen to depths up to 6 inches below the surface.

3.5.2 Temperature.

3.5.2.1 Operating. The ditcher shall perform as specified herein in any ambient temperature from plus 120° F. to minus 25° F. The ditcher shall start within 3 minutes and shall operate under load conditions within 15 minutes. Engine temperatures shall be maintained within the ranges specified in MIL-STD-1400 and all controls and instruments shall function as specified herein.

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3.5.2.2 Storage. The ditcher shall not be permanently damaged by storage under the following conditions:

Air temperature:
High - 155° F. Time period, 4 hours.
Low - Minus 65° F. Time period, 12 hours.

3.5.3 Slopes. The ditcher shall negotiate side slopes of 17.5 percent and longitudinal slopes of 27 percent without any wheel of the ditcher losing contact with the ground.

3.5.4 Durability. The ditcher shall perform all operations specified herein without mechanical adjustment or parts replacement except as specified herein and without malfunction or breakage of structural, mechanical, or electrical components (except lamp bulbs or sealed beam lamps).

3.5.5 Fording. The ditcher shall ford 36-inch-deep streams without damage to the ditcher or its components. The ditcher engine shall be capable of starting and stopping while fording 36-inch-deep streams. After testing as specified in 4.6.2.18, increase of water in enclosures containing lubricants shall not be in excess of 1.5 percent.

3.5.6 Rain. The ditcher shall start within 2 minutes after exposure to a rain of 1/4 gallon per minute per square foot of surface impinging downward on the ditcher while the rain is driven by a 30 mile per hour wind. The vehicle shall operate without malfunction of any electrical component, gage, or instrument. Water in the air intake filter including adsorbed and absorbed water shall not exceed 8 percent of the fluid capacity of the sediment chamber. There shall be no evidence of water leakage into the engine crankcase sump, engine exhaust manifold, turbocharger exhaust (when applicable), transfer case, or fuel tank.

3.5.7 Transportability. The overall dimensions of the ditcher shall be reducible to meet Berne International requirements (see figure 1). Disassembly and reassembly shall each be accomplished in not more than 12 man-hours. Cutting and welding are not permitted. The overall height shall be reducible for transport to 126 inches (table I) without disassembly except that the lower boom support points may be disconnected if necessary. When required, disassembly and reassembly shall each be accomplished in not more than 2 man-hours.

3.5.8 Convoy. The ditcher shall be capable of maintaining an average speed of 20 mph for 3 continuous hours when tested as specified in 4.6.2.20, without breakage, permanent deformation or malfunction of any component, oil temperatures above 250° F., or evidence of inadequate lubrication.

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3.6 Safety. The safety characteristics of the ditcher shall meet the following minimum requirements:

(a) Eliminate concealed danger created by the supplier's design.
(b) Eliminate hazards to personnel and to property.
(c) Result in furnishing material having a correct safety factor.
(d) Provide safety and protective components which do not impair the operating functional requirements. Where it is not possible to preclude the existence of an identified or potential hazard, means for detection of the hazard and generation of a warning signal shall be provided or the supplier shall provide warnings of special procedures to be followed.
(e) Provide correct guarding, enclosing, and insulating features for exposed components and systems which are subject to high temperatures, high pressures, electric actuation, or which are inherently hazardous.
(f) Eliminate nonfunctional sharp edges, projecting points, and excessive length of fasteners.
(g) Provide nonskid footing on all walk and step surfaces.

3.7 Ease of maintenance. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement. Items such as connectors, plugs, and valves shall be standardized as to type, and number of sizes shall be held to a minimum. Drainage of coolant and lubricants except from planetary axle hubs shall be to the ground without draining on or over any part of the ditcher. When drains are located over other parts of the ditcher, integral tubes, pipes, or other means shall be used to convey the coolant or lubricant from the drain to the ground, except that tubes, pipes, or other extensions shall not be required to drain the engine oil. Drains shall be accessible without the removal or adjustment of accessories or parts other than access opening covers. Each operation in the following maintenance operations list shall be accomplished by one man, except as noted, in not more than the time specified, using common tools and special tools furnished with the ditcher:

Maintenance operations list

(a) Remove, replace and adjust engine driven belts - 1 hour.
(b) Remove and replace alternator and adjust belt tension - 1/2 hour.
(c) Remove and replace alternator regulator - 1/2 hour.
(d) Remove and replace power-assist steering hoses - 1 hour.
(e) Remove and replace all filters, screens, and strainers in all hydraulic systems - 2-1/2 hours.
(f) Remove and replace engine coolant system hoses - 1 hour.

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(g) Adjust conveyor belt tension and end pulley alinement - 1/2 hour.
(h) Perform daily servicing and maintenance - 1 hour.
(i) Adjust overload-release mechanism - 1/2 hour.
(j) Remove and replace all teeth from the digging buckets - 8 man-hours (see 3.13.2.1).

3.7.1 Fittings. Disconnect plugs, receptacles, and multiple line connectors in the electrical system and reusable fittings in the pneumatic system shall be utilized to permit component removal and replacement. Reusable straight thread swivel fittings in sizes up to but not including 1-inch nominal size and standard SAE 4-bolt, split-flange, reusable-type fittings in 1-inch nominal size and larger shall be utilized for hose and tubing in the hydraulic system. SAE fittings shall conform to SAE J514 or J516, 37-degree flare. All hydraulic hose fittings shall be suitable for use with bulk hose. All disconnect points shall be clearly identified in the manual.

3.7.2 Frequency of maintenance. The ditcher shall operate for not less than 100 hours with variable loading as specified herein without adjustments other than the supplier's recommended normal scheduled adjustments as established by a maintenance schedule prepared and submitted by the supplier prior to test, and with no parts replacement other than filters, bucket teeth, and conveyor belt lacing.

3.8 Engine. The engine, including all systems, components, accessories, and auxiliaries, shall conform to MIL-E-11276 including provisions for prequalification, except as otherwise specified herein. The engine shall conform to the current Department of HEW Standards for Exhaust Emissions, Fuel Evaporative Emissions and Smoke Emissions, applicable to 1970 and later vehicles and engines, as applicable.

3.8.1 Power and speed rating. The power and speed rating of the engine shall oe such that operation of the ditcher under any of the operational requirements and conditions specified herein will not require horsepower in excess of the continuous horsepower rating of the engine for the applicable governed speed as established by the Qualified Products List, QPL-11276.

3.8.2 Cooling system. The radiator filler neck shall conform to SAE J633 or J868. Coolant hose shall conform to ZZ-H-428, type I or II, grade A, and hose clamps shall conform to SAE J536a, type F, style 4.

3.8.3 Fuel system. The fuel system shall include tank(s) with a capacity sufficient for not less than 8 hours operation in any operating condition specified herein. Fuel capacity shall be based on 85 percent of the fuel consumption rate obtained during conduct of tests in Test

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Method 2000 of MIL-STD-1400, at the applicable horsepower and governed speed. The tank(s) shall: (1) conform to MIL-T-46786, type III or IV, class 2, conforming to table I and with a sending unit for the fuel gage; (2) conform to the requirements of the DOT Motor Carrier Safety Regulations for side-mounted diesel fuel tanks; (3) include a balance line when multiple tanks are furnished; (4) have the engine fuel intake line near the longitudinal center of each tank to preclude starving of the engine on grades and slopes; (5) have a cap, fillpipe, and captive-chained mesh strainer which shall permit filling the tanks at a rate of not less than 20 gallons per minute; and (6) have the fillpipe located so that the tank can be filled from a 5-gallon can conforming to MIL-C-1283. The fuel tank(s) shall be installed on the left side of the ditcher. A steel plate(s) shall be provided under the fuel tank(s) with clearance between the plate(s) and the tank(s) to prevent puncturing by terrain obstacles.

3.8.4 Air cleaner. An air cleaner conforming to MIL-A-52363, class II, and a restriction indicator conforming to MS53063-3 shall be provided. When the air cleaner is mounted with the axis horizontal, an automatic dump valve shall be provided on the primary separator to discharge coarse particles. The gage shall be clearly identified as the air filter gage and shall be visible from the operator's seat.

3.8.5 Exhaust system. The exhaust system shall be located so that removal for shipping shall not be required. The system shall prevent a carbon monoxide concentration in excess of 0.02 percent at the operator's station, and shall provide protection from the entrance of wind-driven rain (see 3.5.6). The exhaust outlet(s) shall be on the right side of the ditcher and directed away from the centerline. The muffler shall conform to MIL-M-52590.

3.8.6 Cranking system. The cranking system shall be of the electric type in accordance with MIL-E-11276, and shall be furnished, or approved, by the engine manufacturer for this application. The starter shall be solenoid operated from the ditcher cab.

3.8.7 Engine housing. An engine housing shall be furnished except that publications compartment and padlocking means are not required, and spring-action panel latches are acceptable.

3.9 Electromagnetic interference. The electromagnetic interference of the ditcher shall be in accordance with SAE J551. Radio suppression of the starter motor and switches associated with starting is not required.

3.10 4 by 4 drive train. The ditcher shall include components to provide 4-wheel drive. These components shall include torque converter, full torque power shift transmission, ditcher machinery PTO(s), transfer case,

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drive shafts with universal joints, planetary-type front steer-drive and rear drive axles, and wheel and tire assemblies. Drive train components shall be installed in accordance with the component manufacturer's recommendations. Drive train fluid and lubricant temperatures shall not exceed 250° F. in any ambient temperature when the ditcher is driven at sustained speeds on smooth highways and when subjected to all operating conditions specified herein. The torque and horsepower rating of each drive train component shall be not less than 150 percent of the maximum torque developed by that portion of the drive train between the engine and that component.

3.10.1 Torque converter and main transmission. The torque converter and main transmission shall absorb maximum stall operation for a minimum of 30 seconds without permanent deformation, damage, or exceeding the fluid temperature limit specified in 3.10, with an initial fluid temperature of not less than 180° F. in an ambient temperature of not lower than 30° F. The torque converter shall be provided with a manual or automatic lockup clutch for direct mechanical drive. The main transmission shall have not less than five forward and one reverse gear ratios. All transmission gears shall be constant mesh. Shifting between gear ratios shall be accomplished by engagement of multiple-disk clutches which shall be actuated by annular hydraulic pistons. The transmission shall provide for shifting under full engine power through all successive forward and reverse transmission gear ratios. The transmission shall provide controlled application of these clutches to provide smooth shifting without slippage which could damage the clutches. Inhibitor mechanisms shall be incorporated in the transmission to prevent down shifts or shifts into reverse that could result in damage to the transmission and engine. Means shall be provided to prevent engine starting from the driver's seat in other than neutral position of the transmission. A heat exchanger shall be provided to stabilize the oil temperatures of the torque converter and the main transmission. The heat exchanger pressure drop shall not exceed 30 psi under maximum flow conditions. The oil shall be filtered by full-flow filter(s) having a replaceable element(s), rated at not greater than 25 micron filtration.

3.10.2 Ditcher machinery (PTO(s)). The ditcher machinery PTO(s) shall provide transfer of engine power to the bucket line drive and a source of power for the crowd mechanism while digging to meet the performance specified herein. When bucket line drive is through transmissions and gear boxes other than an auxiliary truck-type transmission, all bearings shall be pressure lubricated with filtered lubricant; and the alinement of shafts shall be assured by line boring of cases and covers.

3.10.3 Transfer case. Transfer case with interaxle differential shall be furnished in the drive train after the power shift transmission. The transfer case shall mechanically drive the front and rear axles without

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disconnect. The interaxle differential shall proportion torque to the front and to the rear axle drives and shall incorporate a differential lock of either the automatic positive binding type or a driver-selected, power-operated, mechanical lockup type. When the driver-selected lockup is furnished, the control shall be installed on the instrument panel and shall be equipped with a red warning light to indicate lockup engagement.

3.10.4 Universal joints and drive shafts. Drive shafts with paired universal joints shall be furnished in the drive train. Universal joint trunnion bearings shall be sealed with elastomer materials. The universal joint angles encountered during all operations shall be within the limits recommended by the transmission and universal joint manufacturers. On two-piece sliding spline drive shafts, positive means shall be provided to insure correct reassembly and realinement of the drive shaft ends for servicing or parts replacement. Universal joints on the drive shafts to the axles shall be interchangeable and of the same make and model. Each complete drive shaft shall be dynamically balanced at maximum operating speed.

3.10.5 Planetary axles. The front axle shall be of the steering and driving type. The front axle shall be mounted to the frame by either the axle center pivot method or cradle method. The center pivot or cradle mount shall permit oscillation of not less than 16 inches total wheel travel measured at the track center of each wheel. In addition, the axles shall perform as specified in 3.1.1 and 3.5 without damage or deformation. The steering components of the front axle shall permit fine toe-in adjustment and the steering geometry of caster and camber shall be set for optimum handling characteristics and tire life. Elastic axle stops shall cushion contact at full oscillation. A differential lock for the rear axle shall be provided. The lock shall be manually controlled by a switch or lever on the instrument panel, including a warning light adjacent to the control indicating the locked condition.

3.10.6 Wheels, rims, and tires. The ditcher shall be mounted on four single tire and wheel assemblies, all of the same size and ply rating. Tires shall be of the tubeless type, shall have traction lug or rock lug tread design, and shall be in accordance with ZZ-T-1083; except size, tire pressure, and load rating shall be suitable for the application. The supplier shall furnish a certification from the tire manufacturer that the tires furnished shall have a minimum load rating for normal tire life under the conditions specified. Tires shall include commercial antiozonant compounds, and shall be not less than 100-level quality. Tires, rims, valves, and caps shall be of American manufacture and shall conform to and be installed in accordance with the T&RA Yearbook. Tire load capacities shall

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not exceed the tire ratings in the T&RA Yearbook or approved by T&RA for 30 mph travel speed and for 0 mph (static rating) while digging. The recommended operating tire pressure shall not exceed 45 psi. Tires shall be free of major deficiencies when inspected in accordance with MIL-STD-1224.

3.11 Hydraulic system. The hydraulic system shall include circuits to operate and control: (a) the crowd speed; (b) boom hoist; (c) spoil conveyor shift; (d) the conveyor belt drive; and (e) the power-assist steering. The hydraulic pump pressures shall not exceed 2,000 psi. The hydraulic system shall operate on oil conforming to MIL-L-2104, grade 10, or MIL-L-10295 for low-temperature operations. All component materials shall be compatible with these oils and shall operate as specified herein throughout an oil operating range of minus 25° F. to plus 220° F. The temperature rise of the oil in any portion of the hydraulic system shall not exceed 100° F. at any ambient temperature of 50° F. and above during all operations specified herein. The pressure circuits of the hydraulic system shall withstand pressure equal to twice the system working pressure without evidence of leakage, rupture, permanent deformation, or damage. All hydraulic circuits shall include a pressure-relief system which shall be actuated automatically when the maximum operating pressure is exceeded. System components shall be protected against sudden back-pressure surges caused by ditching operations. The hydraulic system shall have full-flow oil filters accessible for cleaning, and the system shall be protected against the entrance of dirt or other foreign material. The filter elements shall conform to MS35802, and the filter housings shall accommodate cover gaskets conforming to MS35802. The MS number and applicable class of the element shall be marked on the surface of the housing. Ports in sizes -4 through -12 shall conform to SAE J514, straight thread "O"-ring boss. Ports in sizes -16 through -32 shall conform to SAE J518, 4-bolt, split-flange type. The system shall maintain a full supply of oil in pumps and suction lines during any operations and conditions specified herein, including start-up after periods of parking or storage. The hydraulic reservoir shall be located on the right side of the ditcher. The temperature of the hydraulic oil in the reservoir shall not exceed 100° F. above ambient during all operations specified herein when ambient temperatures are 90° F. and higher. Inlet ports for all hydraulic pumps shall be lower than the oil surface in the reservoir at the "fill" or "add" level at start-up; or ports and pump chambers shall be flooded by a positive means prior to start-up.

3.11.1 Hydraulic piston rods and control valves. Sliding surfaces shall be plated in accordance with the manufacturer's standard commercial practice with hard chromium and a wall thickness of not less than (a) 0.005 inch for piston rods, (b) 0.0005 inch for control valves. Before being plated, the surfaces shall be hardened to 50 to 55 Rockwell C. After plating, maximum surface roughness shall be 16 microinches, rms. Special process plating may be proposed and used when approved by the contracting officer, provided

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the proposed plating furnishes corrosion resistance equal to or better than the plating required when exposed to salt spray in accordance with ASTM B117 for a period of 30 hours. All control valves shall be hydrostatically balanced.

3.11.2 Valve banks. Valve banks shall be provided with protection against mud, dust, gravel, and water.

3.11.3 Hydraulic lines. Hose shall be used where flexing and vibration may contribute to failure. All other lines shall be of tubing as specified herein. Line routing shall be simple, and provide protection and ease of maintenance. Provisions shall be made to prevent damage of the lines due to chafing. The hose installation, such as angle of fittings and location, shall cause no stress concentration on the hose at the fitting and hose shall be supported when necessary to eliminate sagging. Bulkhead tube fittings shall be used wherever hydraulic lines pass through bulkheads. All lines other than suction lines shall be considered pressure lines.

3.11.4 Pressure hose. Pressure hose and pressure hose assemblies shall conform to MS39317 or MS39318 as applicable. The velocity of oil in all pressure hoses shall not exceed 20 feet per second.

3.11.5 Suction hose. Suction hose conforming to SAE J517, 100R4, shall be used in the hydraulic suction line between the reservoir and the pump wherever vibration may cause failure of a tube and wherever it may simplify the installation. The suction hose size shall be such that the minimum pump inlet pressure will be not less than 7 inches of mercury vacuum. The hose shall have a minimum working pressure of 50 psi and a proof pressure of 100 psi. The hose shall withstand an outside pressure of 15 pounds per square inch absolute (psia) with an inside vacuum of 7 inches of mercury without permanent deformation or damage. The hose change in length shall not exceed plus 2 percent, minus 4 percent when pressurized from 0.0 psi to 50 psi. The hose shall be installed to eliminate stress concentration and shall be free of kinks.

3.11.6 Pressure tubing. Pressure tubing shall be in accordance with SAE J524 or SAE J525. All bends shall be smooth without flattening, kinking, and wrinkling of the tube. The term "without flattening" is defined as follows: Throughout the bend, the maximum diameter minus the minimum diameter shall not exceed 15 percent of the tube nominal diameter.

3.11.7 Pressure tube fittings. Unless otherwise specified herein, fittings in sizes -4 through -12 shall be in accordance with MS39321, except jump size 37 degree flare tees may be used. Tube fittings in sizes -16 through -32 shall contain a female boss into which the tube shall be brazed or welded.

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3.11.8 Hydraulic pump. Gear and vane hydraulic pumps shall conform to MIL-P-52675, except for qualification and identification marking.

3.11.9 Reservoir. The reservoir shall conform to MIL-R-52602, except that the filling shall be accomplished through a filler neck and excepting materials, suction port, return port, air vent, and instruction plate requirements. The filler neck shall have a 3-inch minimum opening and shall contain a strainer retained in place and of sufficient size to strain, by gravity, oil at 70° F. at a rate of 10 gallons per minute. The filler neck shall consist of a filler tube extending to the "FULL" oil level. The top of the filler neck shall extend above the top of the reservoir. Provision shall be made in the filler cap so that a maximum of the first two turns of the filler cap shall release the stored pressure in the reservoir prior to complete filler cap disengagement. The filler neck opening shall be located so as to permit filling with a 5-gallon oil container. The filler cap shall be cast metal.

3.11.10 Hydraulic motors. The hydraulic motors shall be of positive displacement type and shall be mounted so that the motor shaft is subjected to torque only. When component testing is required, the vane or gear motor shall operate continuously without malfunction when tested as specified in 4.6.2.35. Each motor shall withstand a minimum proof pressure of 3,000 psi without permanent deformation, damage, or leakage. The overall efficiency of each motor when operating between 750 psi and 2,000 psi with sufficient oil at a temperature of 150° F. plus or minus 10° F. shall be not less than 70 percent (mechanical output/hydraulic input). There shall be no external leakage such as wetting of the mating portion of the motors or slinging or dripping of oil from the shafts. Port connections shall be large enough to limit the oil velocity to a maximum of 20 feet per second and shall conform to SAE J514. Each motor shall be capable of operating against a back pressure of 500 psi throughout all operating conditions specified herein.

3.11.11 Cylinders. All welding and brazing to the cylinder within the zone of piston operation shall be done prior to final machine honing or rolling of the cylinder. Cylinders shall be located so as to provide for ease of maintenance, replacement, and protection from rocks and any other detrimental material. All hydraulic cylinders shall operate throughout the entire range of stroke and applications without interference or contact by any rigid components of the ditcher. All cylinder pivot points shall be equipped with grease fittings. Drift of each cylinder piston when supporting a load generating 2,000 psi minimum hydraulic pressure shall not exceed 1.00 inch per hour at an oil temperature of 150° F. plus or minus 10° F. Rod packing leakage may wet the rod but shall not collect on the rod packing retainer or rod end of the cylinder.

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3.12 Crowd. Crowd motion of the ditcher shall be accomplished with hydraulic power. The speed shall be infinitely variable from zero to 20 feet per minute when the ditcher is digging in level ground at a depth of 4 feet in undisturbed CL-type soil as defined by the WES Unified Soil Classification System. With the ditcher not digging, the crowd drive shall propel the ditcher on level ground at speeds up to 28 feet per minute. The crowd system shall automatically prevent the ditcher from accelerating beyond a set crowding speed when digging on a downgrade and when moving down the grade in reverse crowd at any crowding speed. The range of crowding speeds shall be obtainable both in forward and reverse directions independent of the bucket-line speed. Any crowding speed change shall not interrupt sustained forward movement of the ditcher or require stopping of the bucket-line. When the hydraulic crowd pump is used to power other components, its capacity shall be such that the simultaneous operation of any other hydraulic components shall not reduce the crowd speeds in the range of 0 to 14 feet per minute by more than 5 percent below speeds set with only the crowd mechanism operating.

3.13 Digging mechanism. The digging mechanism shall include a bucket-line drive, digging-boom assembly, spoil conveyor, the attachment and support components between the digging boom and the carrier, and the follow-up scraper.

3.13.1 Bucket-line drive. Power from the transmission system to the bucket-line shall be transmitted through shafts, gears, chains, a combination of gears and chains, or hydraulically. The alinement of the power drive between transmission and boom shall be maintained automatically throughout the full digging depth range. When power transmission chains are used they shall conform to MIL-STD-421. Adjusting nuts or other means shall be provided for adjusting and maintaining the tension of the power transmission chains. Pins, keys, or other positive means shall be provided to prevent change in the chain tension adjustment during operation of the ditcher.

3.13.1.1 Overload safety release. The overload mechanism shall be located in the bucket-drive power train between the transmission system and the bucket-line. When the bucket-line has been freed of the underground obstructions, the overload mechanism shall automatically resume full torque transmission. When a mechanical overload release mechanism is provided, it shall consist of (a) a brake drum or drums and internal brake shoes with a means for adjusting the amount of transmitted torque or (b) a multiple-disk type clutch provided it is protected from deterioration due to weather, storage, and environmental conditions specified herein. The materials used shall withstand the performance and operational requirements specified in 3.1.1 and 3.5 without adverse effects to the function of the mechanism. The

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release mechanism shall, without slipping and under all conditions specified herein, transmit sufficient torque to develop a bucket-line pull of not less than 12,000 pounds measured along its line of travel. The release point in excess of the required torque transmission shall be determined by the supplier and the mechanism shall be adjusted accordingly, not to exceed a line pull of 14,500 pounds.

3.13.2 Digging-boom assembly. The digging boom shall be of the vertical type with articulated buckets discharging to the rear over the head-shaft. The buckets shall be self-cleaning and the spoil shall be ejected from the buckets by positive action of a wiper plate that is activated by the preceding link. The bucket-line shall have not less than three digging speeds including one of 160 feet per minute plus or minus 20 percent, another of 250 feet per minute plus or minus 15 percent, and a high speed of not less than 350 feet per minute, all at governed engine speed. The bucket-line shall be provided with a means for reversing the direction of travel to unhook itself from underground obstructions. Replaceable covers over the bucket-line on both sides, the front, and the top over the head shaft shall be provided and shall be bolted to a supporting framework.

3.13.2.1 Bucket-line. The bucket-line including buckets, wiper plates, and chain shall conform to the drawings listed on Data List 13201E3200. Cutting teeth shall be installed and shall provide a ditch width of 24-1/2 inches with new teeth, and a nominal cutting width of 24 inches. The tooth arrangement on any two contiguous buckets shall form a pattern providing full coverage of the digging width. The cutting teeth shall be of alloy steel hardened over the wearing surface to 46 to 52 Rockwell C, or surfaced with abrasion-resistant steel. Cutting teeth shall withstand the abrasion caused by ditching for not less than 50 hours in any type of soil as specified in 3.5.1. The complete set of teeth on the bucket-line shall be replaceable in 4 hours by two men using common tools and, when applicable, special tools furnished with the ditcher. Tooth attachment to the buckets shall prevent loss of teeth during ditching operations and during travel. Each mounted tooth and each tooth pocket in the buckets shall not suffer permanent deformation when tested as specified in 4.6.2.3. When in digging position, the clearance between the edge of the tooth on the digging face of the buckets and any part of the ditcher shall be not less than 10 inches on a line parallel to the longitudinal centerline of the ditcher up to the point of discharge to the conveyor.

3.13.2.1.1 Drawings. The drawings forming a part of this specification are engineering design drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the supplier or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

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3.13.2.2 Head shaft. The bucket-line head shaft shall be steel alloy shafting not less than 3-3/8 inches in diameter between bearings, and shall run on antifriction bearings.

3.13.2.3 Sprockets. The head-shaft and foot-shaft sprockets shall be in accordance with the drawings listed on Data List 13201E3200.

3.13.2.4 Foot shaft and adjustment. The foot shaft shall be not less than 2-5/8 inches in diameter, and mounted in the digging boom with provision for removal, replacement, and internal lubrication. The take-up adjustment for bucket-line tension shall be braced to withstand without permanent distortion the dynamic stresses caused by digging and shall provide not less than 4 inches adjustment.

3.13.2.5 Foot-shaft bearings. The foot shaft shall be equipped with bronze bushings conforming to SAE 64 bronze and shall be provided with seals which shall prevent the entrance of soil into the bearing area during all ditching operations specified herein.

3.13.3 Spoil conveyor. The ditcher shall be furnished with a spoil conveyor having a rubber belt 24 inches plus 0 inch and minus 1/2 inch in width.

3.13.3.1 Conveyor belt. The conveyor belt shall conform to MS39281 except for:

(a) Width of belt (see 3.13.3).
(b) Profile of cleats is optional.
(c) Base width of cleats, 1-1/2 inches.
(d) Distance between guide strip notches is optional, suitable for 7-1/2-inch D pulley.

3.13.3.1.1 Mildew treatment. Prior to being coated with rubber, the belt fabric shall be mildew-proofed so that the change in breaking strength will not exceed 20 percent when tested as specified in FED. TEST METHOD STD. No. 191, method 5751.

3.13.3.1.2 Belt lacing. The belt lacing shall be of a type and size compatible with the thickness, tension, reversible direction, and speed of the conveyor belt. The lacing shall withstand the tension and abrasion caused by digging for not less than 50 hours in any type of soil specified in 3.5.1.

3.13.3.2 Conveyor operation and controls. The conveyor shall be adjustable as to point of discharge, extendable to each side, and reversible. A take-up device located in a protected, accessible position shall be provided

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for alining and adjusting tension of the belt. A gage or simple measuring device for checking the alinement of the end pulleys shall be furnished and mounted on the ditcher. The checking device shall enable the operator to determine that the axes of the end pulleys are at right angles with the line of belt travel, and to adjust the end pulleys to the required alinement. The conveyor shall be not more than 8 feet in overall length measured on a straight line from the outer periphery of one end pulley to the other. The diameter of the end pulleys shall be not less than 7-1/2 inches. The conveyor shall be driven by hydraulically powered gear boxes located at each end pulley or by direct drive hydraulic motors. Each end pulley shall have a central groove to accommodate the C-section guide on the belt, and the sides of the groove shall have the same taper as the guide strip. The top of the groove shall be between 1/8 inch and 1/4 inch wider than the base of the guide strip; and a clearance of not less than 1/2 inch shall be provided between the top of the guide strip and the bottom of the groove. Not less than four idler pulleys shall be provided. Rubber collars or rings on the idler pulleys shall be provided and shall cushion the shock loading on the belt while ditching in the soils specified in 3.5.1. The hydraulic motors shall drive the conveyor belt at infinitely variable speeds from zero to not less than 650 feet per minute. The travel direction of the conveyor belt shall be instantly reversible. Speed and direction shall be controlled from the operator's station. The hydraulic circuit driving the discharge conveyor motors shall be independent of any other hydraulic circuits. The conveyor shall be shiftable from one side of the ditcher to the other by means of power from the hydraulic system and shall be controlled from the operator's station. The shiftable distance to either side shall be not less than 1 foot, 10 inches measured from the centerline of the ditching machine. With the ditcher digging as specified in 3.1.1, the conveyor shall deposit the spoil at a sufficient distance to prevent spillage back into the excavation; and the deposited spoil shall not contact the conveyor. The conveyor shall be provided with suitable scrapers for keeping the driven side of the belt and the end pulleys in a clean condition.

3.13.3.3 Flashing. Flashing shall be provided and shall be 1/4-inch-thick, 5 percent wax impregnated, abrasion resistant, natural rubber compound of 2,500 to 3,000 tensile strength, 500 percent elongation, 40 to 50 durometer.

3.13.4 Boom support. The digging boom shall be held in its various digging and traveling positions at all times through both upper and lower connecting points of pivoted arms or rollers in guides. The boom shall be raised and lowered by hydraulic rams with cable connection between the main hydraulic ram and digging boom to permit flexible upward movement of the digging boom under shock conditions.

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3.13.4.1 Digging position. Hydraulic valve controls at the operating station shall permit the operator to control the depth of digging. While the ditcher is digging, the longitudinal forces between the boom and the chassis shall be transmitted through the mechanical structure and not by hydraulic means.

3.13.4.1.1 Digging-depth indicator. A digging-depth indicator shall be provided and located so as to be visible to the operator. The indicator dial shall be marked to indicate each 1/2 foot of the digging depth range, with whole foot marks numbered to indicate depth of cut in feet. The indicator shall be visible to the operator in his normal operating position.

3.13.4.2 Travel position. The digging boom shall be suspended so that it can be raised to travel position as specified in table I by the operator working the controls from digging station.

3.13.4.2.1 Boom lock. A positive lock to prevent upward and downward movement of the boom during travel shall be provided. The lock shall include pin connection(s) through the boom guide frame and the boom. A mechanical keeper device shall be provided to secure the lockpin in place. The positive lock shall function independently of the hydraulic system.

3.13.4.3 Transport. The digging boom shall be suspended so that when the ditcher is being transported, the overall height of the ditcher shall conform to table I when the boom is lowered to the surface of the carrier (see 3.5.7).

3.13.5 Follow-up scraper. The ditcher shall be provided with a follow-up scraper which will clean loose soil spillage from the bottom of the trench. The scraper shall be hinged to swing free of the ground when not in use and shall be raised and lowered hydraulically. The scraper hydraulic control shall be equipped with a pressure relief device as specified in 3.11.

3.14 Digging controls. All digging controls shall be located within reach of the operator when he is seated at the digging control station on the left side of the ditcher. Ball and socket joints in rod-type control linkages shall be of the adjustable spring-loaded type. Flexible control linkages shall be of rust-resistant materials, watertight, nonlubricated, and sealed to prevent the entrance of dust and water. Photographic plates conforming to MIL-P-514, type I, size as applicable, shall be provided to identify each control and its operating positions. The controls shall be illuminated for night operation.

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3.15 Sound level. The ditcher sound level at the normally seated operator's ear position shall be determined in accordance with SAE J919. Sound level measurements shall be taken under the following stationary and moving conditions:

(a) High idle.
(b) Low idle.
(c) Full throttle with bucket-line running.

Resulting values shall not exceed those listed below:

Range	Frequency Bands (cps)	Maximum Acceptable Noise Level (db re 0.0002 microbar)
(a)	below 75	120
(b)	75 - 150	115
(c)	150 - 300	109
(d)	300 - 600	101
(e)	600 - 1200	93
(f)	1200 - 2400	89
(g)	2400 - 4800	89
(h)	4800 - 9600	91

3.16 Front bumper guards, fenders, and decking. The ditcher shall be equipped with a wrap-around-type, heavy, front bumper fabricated of not less than 1/4-inch steel plate or of equivalent structural steel. The front bumper may also serve as a counterweight. A radiator and headlight safety grille guard of welded steel construction with frame braces shall protect the front of the ditcher from brush and sapling tree damage. A wire mesh or punched metal screen having openings not larger than 1/4-inch shall be mounted between the guard and radiator. Front fenders shall extend over the front tires with a minimum clearance of 12 inches, shall support a weight of 200 pounds applied on a 10-square-inch area, and shall have a nonskid surface. Rear fenders shall be provided and shall be fabricated to withstand the same applied forces as the front fenders. The fenders shall protect the rear lights, oil tanks, cab, and the digging control station from mud and water splashed from the road or thrown off the tires. Steps shall be provided for access to the cab; to the operator's station from the ground, and between the cab and the operator's station. The steps and the decking of the operator's station, steps between the cab and the operator's station, and walkways and standing areas used during maintenance shall have nonskid surfaces.

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3.17 Power-assist steering system. The ditcher shall be equipped with a hydraulic power-assist steering system. A flow control shall provide a relatively constant volume regardless of variations in engine speed. The maximum relief valve setting shall not exceed 2,000 psi. A friction-type, steering-wheel-rim brake shall hold the position of the front wheels during the digging operation and allow the operator to make steering changes by overriding this brake. The clearance between the outer edge of the steering wheel and the nearest projection shall be not less than 2-1/2 inches. The steering column may include a universal joint or flexible coupling near the gearbox. The ditcher shall be steerable in the event of unexpected engine failure and as a towed vehicle with a dead engine. With the engine at idle speed, the minimum power performance of the steering system shall permit rotation of the steering wheel 360 degrees in not more than 5 seconds with a driver's rim pull effort not to exceed 10 pounds when the ditcher is stationary on level pavement.

3.18 Frame. The frame shall provide strength for imposed static and dynamic loads, including reactions from lifting the ditcher. The frame assembly shall provide maximum accessibility to all drive-line components. Two towing lug pad eyes conforming to MS500004 for use on 15-ton trucks shall be installed on the front bumper. Each pad eye shall withstand without damage the imposed forces when the ditcher is pulled by another vehicle capable of exerting up to 36,000 pounds horizontal straight-ahead pull on either eye; and when attached to a V-type towbar conforming to MIL-T-45043, size 2 or 3, and subjected to a 35-ton pull while the ditcher is stationary.

3.19 Brake system. The ditcher shall be equipped with a weatherproof, straight airbrake system that shall comply with the requirements of the California Vehicle Code, Division 12, and the DOT Motor Carrier Safety Regulations and shall be as specified herein. Application of the service brakes when traveling at a speed of 25 mph shall stop the ditcher for repeated stops in a distance of not more than 50 feet when tested as specified in 4.6.2.14. All brake-system lines and components shall be located, installed, and protected so that damage from ground obstructions and debris is effectively prevented and that maximum efficiency and ease of maintenance is obtained.

3.19.1 Compressor. The compressor shall be water-cooled, engine-driven, engine-oil-lubricated with a rating of not less than 12 cfm. The compressor shall have an automatic pressure control with a factory set pressure range of 95 to 120 psi. The compressor shall draw filtered air from the engine air cleaner manifold through a check valve.

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3.19.2 Alcohol evaporator. An alcohol evaporator conforming to MS53067 shall be furnished, including necessary check valves. The evaporator shall mist alcohol into the compressed air system to prevent condensate freezing.

3.19.3 Reservoirs. Wet and dry reservoirs with a combined capacity of not less than 3,600 cubic inches shall be provided. The reservoirs shall conform to SAE J10. Each reservoir shall be equipped with a safety valve set to release at 150 psi and a drain cock. The drain cocks shall be located in a protected position that will permit opening and closing the cocks without necessitating personnel crawling underneath the ditcher except under the front bumper.

3.19.4 Foot-application valve. The foot-operated treadle valve shall be of the high-capacity type. The location of the application valve piping shall prevent condensate draining into the valve.

3.19.5 Front axle combined limiting and quick-release valve. A combined limiting and quick-release valve shall be furnished for front axle brake operation and shall include a control installed on the instrument panel for variation of front axle braking application.

3.19.6 Rear axle relay valve. A quick-release relay valve shall be furnished for rear axle brake operation.

3.19.7 Low-pressure indicators. One flag-type visual and one buzzer-type audible low-air-pressure indicator shall be furnished on each ditcher. The buzzer and the operating swtich shall be as specified in 3.20.

3.19.8 Stoplight switch. The stoplight switch shall be as specified in 3.20.

3.19.9 Service and emergency air couplings. Service and emergency air couplings conforming to MS35746 with spring covers and a manual cutoff cock shall be installed on the front of the ditcher and shall be connected in a manner such that when the carrier is inoperative and is towed by another vehicle, the ditcher's airbrake system shall operate from the towing vehicle.

3.19.10 Brake lines. Main brake lines shall be installed with reusable fittings in accordance with the SAE Handbook. The brake lines shall be of sufficient size to deliver a minimum of 60 psi chamber pressure to the brakes without lag time exceeding 0.4 second.

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3.19.11 Brake drums and shoes. Brake drums shall be of the anti-bellmouth type and the wearing surface material shall be SAE 114 or SAE 115 automotive gray cast iron. Heavy cast or fabricated shoes shall be furnished to resist heat warpage. The shoes shall be lined with a sintered iron-based lining mixture which shall be fused to a steel backing and installed on the shoes by the use of hardened steel bolts. The lining mixture shall conform to KKK-L-370, type II, except as modified for air-brake service.

3.19.12 Emergency stopping system. The emergency stopping system shall conform to the California Vehicle Code.

3.19.13 Parking brake system. The parking brake system shall conform to DOT Motor Carrier Safety Regulation. The emergency stopping system may be used as the parking brake system provided the requirements in the DOT regulation are fulfilled. The system shall provide a braking force to hold the ditcher on grades up to 35 percent.

3.19.14 Manual controls. Two controls shall be furnished for operation of the parking brakes. One control shall be located in the cab and shall be accessible from the driver's seat; the second control shall be located adjacent to and accessible from the digging control station.

3.20 Electrical system. The ditcher shall be equipped with a 24-volt, waterproof, vehicular electrical system for service and blackout night travel operations, for floodlighting the work area at night, for slave starting the engines of other vehicles, and for charging the batteries and starting the engine. Except as specified herein, the electrical system shall be in accordance with the following:

(a) SAE Handbook, Electrical Equipment and Lighting Section.
(b) DOT Motor Carrier Safety Regulations.
(c) MIL-STD-646, except for color coding.
(d) Figures 2 and 3, with the exception of deleting the rear identification lights and the trailer receptacles.

The electrical system shall include the components listed in table II, plus additional items required for a complete system. Item quantities not specified shall be furnished as required.

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Table II. Electrical System and Instruments

Description	Requirement	Quantity
Battery, storage, 12 v, 6 TN	MS35000	-
Holddown frame	MS53046-3	1 for ea 2 batteries
Piano hinge	MS35829-C	-
Alternator and regulator, 35 amp rating	MS39248 and MS39249, MS39250 and MS39251, MS39252 and MS39253, or commercial equivalent with transistor-type regulator.	1 ea
Low-tension cable	SAE J555a	-
Harness insulation	MIL-I-15126 MIL-I-631	-
Loom	SAE J562 oil-proof	-
Cushioned support straps	Commercial	-
Circuit number bands	MS39020	
Friction connectors	MS27142 thru MS27148, and MS33800	-
Connector lubricant	MIL-S-8660	-
Circuit breaker	MS39062	-
Slave receptacle	MS75058	1 ea
Slave cable w/plugs	SAE J555a 240-inch length and MS75057	1 ea
Speedometer and tachometer shafts	SAE J678d	-
Low-air-pressure switch	MS27152	1 ea
Low-air-pressure buzzer	Commercial	1 ea
Air horn	MS51301	1 ea
Panel light	Commercial	3 ea, min.
Warning and indicator light assemblies	MIL-L-3661 MIL-L-3661/7, MIL-L-3661/15	-
Ammeter	Commercial	1 ea
Pressure gage, engine oil	(see 3.21.7)	1 ea

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Table II. Electrical System and Instruments (cont'd)

Description	Requirement	Quantity
Pressure gage, airbrake	(see 3.21.7)	1 ea
Temperature gage, engine coolant	(see 3.21.7)	1 ea
Fuel gage	MS24544	1 ea
Speedometer - Odometer	SAE J678d	1 ea
Tachometer - Hourmeter	SAE J678d	1 ea
Rotary switch	MS39060	-
Toggle switch	MS39061	-
Main light switch	MS51113	1 ea
Dimmer switch	SAE J564	1 ea
Turn signal control	Commercial	1 ea
Hazard warning flasher assembly	DOT Regulations	1 ea
Left side blackout headlight	MS51318	1 ea
Front blackout marker	MS51303	2 ea
Service headlights	Flush mtd type similar to MIL-L-13071	4 ea
Sealed beam units	Trade No. 4880	4 ea
Front turn signal lights	SAE-PI, 4-inch dia, waterproof amber	2 ea
Service, identification, side-marker	MS35423	-
and clearance lights	(see 3.20.5)	-
Blackout side-marker and clearance lights	MS35424	-
Service combination lights	SAE-S-I-T, 4-inch dia, waterproof	2 ea
Blackout combination lights	MS51330	2 ea
Work lights	SAE J598	2 ea
Sealed beam units, flood	Trade No. 4578	2 ea
Reflectors	SAE J594	8 ea

3.20.1 Batteries. The batteries shall be supplied in pairs in sufficient quantity to furnish the required ampere-hour capacity. All lug terminals shall be removable without requiring disassembly of other components. The battery enclosure shall conform to 3.20.7.

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3.20.2 Slaving components. The slave receptacle shall conform to 3.20.9. The slave cable shall be stowed in the toolbox.

3.20.3 Starting system. The starting system shall conform to 3.8.6.

3.20.4 Wiring. The wiring of the ditcher shall be fabricated and installed as recommended in SAE J555a, except as specified herein. Where two or more wires follow the same routing, the wires shall be formed into harnesses and either bound together with tape conforming to MIL-I-15126, type CFT or covered with insulation conforming to MIL-I-631. Battery cables shall satisfy the SAE starter voltage drop for heavy-duty diesel use and current requirements, shall be as short as possible, and shall not be spliced. All wires shall be number coded near each connection following the schematics and the circuit numbering charts. Color coding is not required. Friction retainment connectors shall be lubricated prior to assembly and each connector assembly shall be secure. Removal of major assemblies such as engine, cab, and instrument panel shall be facilitated by disconnecting one or two threaded plugs and receptacles for all 24-volt circuits in each assembly. Terminal-type connectors may be used on the floodlight system.

3.20.5 Lighting. The ditcher shall be equipped with service and blackout lighting as shown in figure 2, with the exception of the rear identification lights, turn signal control, turn signal relay box, and the trailer receptacle; and a single light mounted diagonally may replace a pair of clearance lights at the corners. Floodlights and headlights shall be recessed or guarded for protection from damage drom tree limbs or falling debris. The service lighting shall conform to the DOT Motor Carrier Safety Regulations for large trucks, including required marking on the boom. The blackout lighting system shall function in accordance with the wiring diagram. All light housings shall provide maximum shock and vibration protection by location and through means of resilient mountings. Replacement of bulbs and sealed units shall not disturb permanently installed light housings. The outer or lower headlights shall be for low beam and the inner or higher headlights shall be for high beam. Only the two outer headlights shall be illuminated on low beam, and all headlights shall be illuminated on high beam. The turn signal control shall be installed on the steering column or on the instrument panel. The emergency hazard warning control shall be located on the lower left of the dash and shall not interfere with the installation of the heater control, or may be combined with the turn signal control. The emergency hazard warning flasher assembly shall consist of a 24-volt, 60- to 75-cpm, variable-load, class A flasher; a switch; a red pilot light; and leads to the distribution box

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terminals to flash the lights. The two floodlights shall be located on each side of the ditcher to illuminate the ditch area and the buckets. The circuit shall be protected by a circuit breaker and the floodlights shall be controlled from a toggle switch mounted on the instrument panel near the main lighting switch. Mounting brackets for the pedestal floodlight shall include brush guards. The brackets shall be fabricated or formed from steel not less than 1/4 inch thick and shall provide an aiming adjustment of the light pattern of not less than 10 degrees above and 20 degrees below the centered light beam pattern.

3.20.6 Reflectors. Reflectors shall be furnished and mounted as required by DOT Motor Carrier Safety Regulations. In addition, two amber reflectors shall face to the front, one on each side of the ditcher. These reflectors shall not be mounted on the face of the bumper.

3.20.7 Battery enclosure. A weathertight enclosure of the box type or compartment type shall be provided to contain and to protect the batteries. The enclosure shall (1) have not less than 30 inches ground clearance; (2) hold the batteries in a horizontal plane when the ditcher is on level ground; (3) include drains for removal of water or electrolyte spilled within the container, and gas venting at or near the top of the container; and (4) provide accessibility for servicing and removal of the batteries. The location of the battery enclosure shall afford clear vision of electrolyte level in all cells of the batteries. Sheet metal used to fabricate the enclosure shall be of steel not less than 0.062-inch nominal thickness. Batteries shall be restrained in the enclosure using the components listed in table II. The interior of the enclosure, including holddown hardware, shall be coated with a phenolic- or asphalt-base acid resistant material.

3.20.8 Batteries. Batteries shall be size 6 TN conforming to MS35000. Batteries shall be furnished charged and dry with a separate filling of electrolyte furnished with each battery.

3.20.9 Slave receptable. A receptacle conforming to MS75058 shall be installed near the batteries to permit engine slave starting and battery charging. The mounting location shall allow insertion of the slave cable plug by personnel standing on the ground. A slave cable with plugs shall be furnished in accordance with table II, and stowed in the toolbox.

3.21 Cab. The ditcher shall be equipped with a two-man, open, coupe-type cab having folding windshield, soft top with bows, and side curtains for weather protection. Soft top and curtain material and construction shall be in accordance with MIL-C-13489. Curtains shall extend below the floor or doorsill level. The cab shall provide not less than 200 square

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inches of fresh-air openings in the cowl, including the fresh air intake for the heater. The openings shall include screening and weather sealing and shall provide multiposition adjustment between fully open and fully closed by personnel from their seats. Airflow around the feet of the driver and passenger shall be provided. The cab shall withstand without damage or permanent deformation the stresses and vibration caused by all operations specified herein.

3.21.1 Steps and handholds. Steps and handholds shall be provided on both sides of the cab. Nonskid steps shall be positioned to permit safe cab entry and exit by personnel wearing arctic clothing. The handholds shall be suitable for use by personnel wearing arctic mittens and shall permit safe entry and exit by personnel while holding on with one hand.

3.21.2 Windshield. A folding frame, full-width windshield shall be furnished. The windshield shall be two-piece right and left side and shall be laminated plate glass conforming to MIL-G-3787, class I, type A. The windshield glass shall be mounted and held in its frame by heavy, rubber sealing. The windshield frame shall be pivoted or hinged near or at the bottom to enable the operator to fold the windshield forward and secure it in a level position over the engine compartment. The joint shall be sealed against wind and rain. Positive latches shall secure the windshield in the closed and folded-down positions until released by the operator.

3.21.2.1 Windshield cover. A windshield cover shall be furnished and shall be made of the same material as specified for the cab soft top. The cover shall slip over the windshield when it is folded down forward and shall be secured against loss from vibration or wind. When in place, the windshield cover shall completely cover both sides of the windshield glass.

3.21.2.2 Windshield components. Dual windshield wipers and dual sun visors shall be furnished. Wiper arms and blades shall be of the commercial, heavy-duty, truck and bus type having saddle and pin blade fastening. Wiper arms shall be not less than 16 inches long. Two heavy-duty electric wiper motors with heavy-duty drive shafts and transmissions shall drive the arms. The length of each sun visor shall be not less than 50 percent of the width of each windshield section.

3.21.3 Windows. The side and rear windows shall be of clear acrylic plastic and the window width shall be not less than two-thirds of the available opening.

3.21.4 Rear view mirrors. Two west coast type mirrors shall be furnished. The mounting shall be adjustable and shall prevent vibration from blurring the image of either mirror. Each mirror assembly shall include dual, fore

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and aft brush guards that will protect the mirror heads. The guard shall ratchet or swing from its lock-joint when contacted by an obstruction exerting in excess of 15 pounds on the guard and when necessary for clearance. The mirror assembly shall relock when returned to its original position manually, without requiring any readjustment. The mirror head shall be not less than 6 by 16 inch nominal size and constructed of heavy sheet or cast aluminum. Sheet aluminum shall be not less than 0.032 inch in thickness. The head shall house a polished plate glass mirror of not less than 1/4-inch thickness, shock protected in a waterproof seal. The mirror glass shall be replaceable. The mirror head shall be mounted by studs at the top and bottom of the head. Fasteners used in the mirror assembly shall be protected from corrosion, and all nuts used shall be of the elastic type that shall not loosen in service.

3.21.5 Seating. The driver's seat shall be suspended on an air bag or torsion springs and heavy-duty shock absorbers which shall be adjustable to the weight of the driver. The driver's seat shall be in accordance with SAE J899 except that the width shall be not less than 18 inches. In its central position, the seat shall be located so that the operator has maximum unobstructed visibility of both the road and all gages, indicators and lights, and has access to all travel controls. Adjustment shall be made without tools. The seat cushion and back shall be padded with foam rubber or polyurethane. The covering material shall be resistant to abrasion and moisture and shall not be adversely affected by the extremes of temperature specified herein. A portable seat with a back support shall be furnished for use as a second seat in the cab and as a seat for the operator at the ditching control station. The weight of the portable seat assembly shall not exceed 40 pounds. The seat position at the ditching controls shall provide the operator with access to the controls and a clear view of the digging operation. For traveling, the portable seat shall be positioned in the cab on the right side facing forward. Padding and covering for the seat and back shall be the same as required for the driver's seat. No springs or springing mechanism shall support the portable seat. The seat shall be securely held down in either location and shall not require tools for mounting and removal. Seat belts conforming to JJ-B-185 shall be furnished for each seat and shall be of the metal-to-metal, quick-release-buckle type. Seat-belt anchors shall conform to SAE J787 for seat belts attached to the seat frame.

3.21.6 Heater and defroster. An engine coolant, fresh-air-type heater with defroster shall be furnished, with a rating of not less than 30,000 Btu/hr. The heater air intake shall include screening and weather sealing. The heated air shall be evenly distributed within the compartment by means of built-in ducts. Windshield defroster and toe board outlets shall be of

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the louver type extending not less than 60 percent of space width to be heated. Controls shall be provided for variable selection of fresh air volume, fresh or recirculated air, floor or defroster outlet air, blower speed, and heater temperature (coolant volume). The controls shall be grouped in the same general location, primarily accessible to the driver. The heater installation shall be supplied with drain valves and, if necessary, air lock bleed valves. All exposed heater components and piping shall be protected against damage and shall be insulated.

3.21.7 Cab controls and dash. All traveling controls for the ditcher shall be located within the cab and shall be within reach of the operator from his seat. The driving operation of the ditcher shall be in the normal automotive manner. The instrument panel components and arrangement shall be in accordance with SAE J680, using the components specified in 3.20. Rubber shock-mounting of the instrument panel, or of individual instruments and indicators, shall be furnished. The panel shall include all primary diesel instruments, indicators, and controls and at least the following secondary instruments, indicators, and controls:

(a) Coolant temperature and oil-pressure warning lights.
(b) Brake system controls.
(c) Heater and defroster controls.

All instruments, other than military, shall be commercial, waterproof, fluid-operated type with normal operating range and danger ranges marked on the face in addition to numerical values. The instruments shall be of such quality and accuracy and so mounted as to function under all conditions specified herein. Plates conforming to MIL-P-514, type III, photographic plate, size as applicable, shall identify each component, and the plates shall be illuminated for night operation. Access to all connections shall be provided by removal of the panels from the front. Warning lights shall be installed above the corresponding instrument or indicator. The two green turn-signal flash indicator lights shall be located one on each upper side of the main panel. A dispatch box having dimensions not less than 8 by 6 by 4 inches shall be installed behind the right side of the instrument panel, with the opening through the panel fitted with a hinged, fold-down, watertight door and latch. When cab controls operate linkages of the flexible push-pull type, the flexible linkage shall be of rust-resistant materials, watertight, nonlubricated, and sealed to prevent entrance of dust and water. Ball and socket type joints used on rigid control linkages shall be of the adjustable spring-loaded type.

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3.22 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating.

3.22.1 Lubricants. The ditchers shall operate as specified herein when serviced with military products and antifreeze as listed in table III.

Table III. Service Products

Item application	Specification	Type & Grade
1. Engine oils		
(a) Heavy duty	MIL-L-2104	10, 30, 50
(b) Arctic	MIL-L-10295	One grade
2. Fuel, diesel	VV-F-800	DF-1, DF-2, and DF-A
3. Grease		
(a) Automotive	MIL-G-10924	One grade
(b) High temperature	MIL-G-23827	One grade
4. Hydraulic-system oil (excluding brakes)		
(a) Normal ambient	MIL-L-2104	10
(b) Low temperature	MIL-L-10295	One grade
5. Lubricating oils		
(a) Universal	MIL-L-2105	75, 80, 90
(b) Arctic	MIL-L-10324	One grade
(c) Instrument	MIL-L-6085	One grade
6. Antifreeze		
(a) Ethylene glycol	O-A-548	I or II

3.22.2 Lubrication fittings. Lubrication fittings shall conform to MS15001, MS15002, or MS15003. Fittings shall be located in a protected position. Fittings shall be accessible to a grease gun conforming to MIL-G-3859 equipped with a hydraulic coupler conforming to MS24207 and a 10-inch flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Engine-housing

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side panels and plates equipped with hand-operated, quick-disconnect fasteners may be removed to provide accessibility to lubrication fittings. Lube locations difficult or awkward to reach shall be extended by tubes or other means to provide easy accessibility.

3.22.3 Pressure-release device. A pressure-release device shall be provided where the use of pressure lubricating equipment will damage grease seals or other parts.

3.22.4 Enclosures with reservoir of liquid lubricant. Enclosures such as gear cases, transmission housings, and engine crankcases which contain a reservoir of liquid lubricants for lubrication of the parts enclosed shall be equipped with dipsticks, check plugs of not less than 1/2-inch pipe size, or sight gages to determine the level of the lubricant. Each enclosure shall be equipped with a means of filling the enclosure with lubricant and for draining. The drain shall be fitted with a magnetic drain plug having either tapered pipe threads, sizes conforming to MS49006 or shoulder and gasket, sizes conforming to MS35844. The drain plug shall be located so that removal of the plug will result in complete drainage of the lubricant from the enclosure. Plates equipped with hand-operated, quick-disconnect fasteners may be removed for servicing.

3.22.5 Delivery lubrication service. Unless otherwise preserved for packaging (see 5.2), all components shall be filled to the operating level with winter grades of military oils designated for use in the temperature range of 0° F. to 32° F. Grease lubrication, including lubrication of sealed bearings, shall be with grease conforming to MIL-G-10924 or MIL-G-23827, as applicable. Ditchers shall be tagged in a conspicuous place in the operator's compartment indicating the temperature range and grades of oils and greases used.

3.23 Lifting attachments. The ditcher shall be provided with lifting attachments. The attachments shall enable the ditcher to be lifted in a normal traveling or transport position. The attachments shall be located so that not less than 1 inch clearance is maintained between the slings and all exterior parts of the ditcher. Spreader bars are not permitted. Each attachment shall carry not less than 2-1/2 times the normal load imposed on the attachment. Attachments for multiple slings shall be located so that the attached slings shall converge not more than 19 feet above the lowest extremity of the load. Attachments shall be positioned so that the lifting strain will be in line with the longitudinal axis of the attachment. The inside diameter of the eye shall be not less than 3 inches. Attachments shall be fastened to members which will withstand stresses in the amount and direction of pull specified for the attachment without weld failure or permanent deformation of the ditcher or of the attachment. Where required, members shall be reinforced to meet the requirements specified herein. Lifting attachments may also be used as tiedown attachments when such attachments meet the requirements as specified in 3.24.

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3.24 Tiedown attachments. Permanently affixed attachments that permit fastening the ditcher to the floor or deck of a transportation medium shall be provided. Each attachment shall withstand, without damage to the ditcher or to the attachment, the dynamic loads applied when tested as specified in 4.6.2.10. Attachments shall be rounded and beveled to prevent cutting of the tiedown material. Tiedown attachments may also be used as lifting attachments when such attachments meet the requirements specified in 3.23.

3.25 Identification marking. The ditcher shall be identified in accordance with MIL-STD-130. The marking shall be applied to the ditcher on identification plates conforming to MIL-P-514, type I, photographic plate. A transportation-data plate conforming to MIL-P-514, type III, photographic plate, shall be furnished and shall indicate location and capacity of lifting and tiedown attachments, and of the longitudinal and vertical locations of the center of gravity.

3.26 Instruction plates. The ditchers shall be equipped with instruction plates describing any special or important procedure to be followed in operating and servicing the ditcher and preparation for rail shipment. Plates shall conform to MIL-P-514, type III, photographic plate.

3.27 Treatment and painting. The portions of the ditcher normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, except that commercial cleaning, treating, and prime coating will be acceptable provided they comply with the following:

(a) The commercial paint and application process is equal to or better than the cleaning, pretreatment, and priming required by MIL-T-704.
(b) The commercial paint supplied is compatible with the finish coat as required by MIL-T-704. The finish color shall be No. X24087.
(c) The finished paint thickness will be equal to or greater than the total thickness required by MIL-T-704 for prime and finish coats. Parts subject to temperatures in excess of 450° F. shall be painted in accordance with the supplier's standard practice.
(d) Crankcase sealer - The application of a crankcase sealer as specified in MIL-T-704 on the interior surfaces need not be applied on the cast iron housings of transmission and torque converters provided the interior surfaces are void of casting defects such as blowholes, excessive porosity, cold shuts, hard spots, shrinks, cracks, or other injurious defects.

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(e) Brake protection - Surfaces of steel and iron brake assembly components which are exposed to air shall be protected by treatment conforming to MIL-T-704, type B, except that the finish coat is not required. Bearing surfaces in brake assemblies such as brake shoe anchor pins, anchor pin bearing holes in brake shoes, cylindrical surface of pistons in brake cylinders, and inside diameter of brake cylinders for the length of the piston contact, shall be coated with a film of silicone compound conforming to MIL-C-21567. A film of silicone compound shall be applied to the lip and yoke hole of each dust boot and to the boot lip groove in the outside diameter at the ends of each cylinder.

3.28 Vehicle marking. Registration and other markings shall be applied in accordance with MIL-STD-642. The gross weight shall be stenciled on each side of the ditcher. The tire pressure prescribed by the supplier shall be stenciled on the instrument panel.

3.29 Toolbox. The ditcher shall be provided with a toolbox made of sheet steel 0.063 inch in nominal thickness or with a compartment-type tool container. The toolbox shall have a hinged lid and hasp-type fastener capable of receiving a padlock conforming to MS35647-1. The lid shall open at least 90 degrees from the closed position. The toolbox shall be fastened to the ditcher in a protected position and the box or compartment shall be of a size to accommodate the tools furnished. Provisions shall be made to drain the toolbox in its normal position.

3.30 Items to be furnished with ditchers. No shipment of ditchers shall be made unless repair parts, maintenance tools, and accessories are included with the shipment or unless approval for shipment of the ditchers without such items has been received from the contracting officer.

3.31 Repair parts and maintenance tools. Such repair parts and maintenance tools as are specified shall be furnished. Special tools required for operation, field adjustment, or daily and weekly maintenance of the ditcher shall be furnished with each unit.

3.32 Rifle case. A Government-furnished rifle case with holder assembly FSN 2590-045-9611 (see 6.3) shall be mounted on the ditcher in a location accessible to the operator. The location and method of mounting shall be subject to the approval of the contracting officer prior to installation. All necessary fastenings and fittings shall be furnished by the supplier.

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3.33 Fungus and moisture resistance. The electrical circuitry, including all components and connections except for the following, shall be protected from the effects of moisture and fungus growth by an overall treatment with a varnish conforming to MIL-V-173 or a varnish having a 1 percent copper 8-quinolinolate content (by weight) based on the nonvolatile content of the varnish:

(a) Components or circuit elements which are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or functioning of the varnish.

3.34 Workmanship. All parts, components, and assemblies of the ditcher including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces, and welded parts shall be clean and free from sand, dirt, fins, pits, sprues, loose mill scale, flux, and other harmful extraneous material. External surfaces shall be free of burrs, sharp edges, and corners except where sharp edges and corners are required.

3.34.1 Steel fabrication. Steel used in the fabrication of the ditchers shall provide original quality surface finish and shall be free from kinks and sharp bends. Burned surfaces of flame-cut material shall be free of burrs, sharp edges, and slag. All bends of a major character shall be made with metal dies or fixtures to insure uniformity of size and shape.

3.34.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with commercial practice. All bolts, nuts, and screws shall be correctly torqued.

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3.34.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall completely fill the holes. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.34.4 Welders and welding.

3.34.4.1 Welders. Before assigning any welder to manual welding work covered by this specification, the supplier shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operation to be performed and that such qualification is effective as defined by the particular code:

Standard Qualification Procedure of the American Welding Society.
Welding Qualification of the ASME Boiler and Pressure Vessel Code.

3.34.4.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loadings. Welded parts and assemblies shall be free from cracks and other imperfections that may reduce the effectiveness of the weld or of the part. All weldments shall be free from slag, flux, weld spatter, and other impurities.

3.34.5 Machine work. Tolerances and gages for metal fits shall conform to the limitations specified herein and on the applicable drawings.

3.34.6 Aluminum fabrication. Fabrication of aluminum shall be done by methods that will not cause injury to the metal. Care shall be used in handling to avoid scratching and damage to the edges. Finished edges shall be smooth and free from laminations, kinks, bends, winds, and scratches. Corners shall be square and true. Bends shall be made with metal dies or fixtures to assure uniformity of size and shape. Sharp instruments shall not be used for marking bend lines.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Throughout the testing period the supplier shall maintain complete and current test data which shall be made available to the Government Representative(s) for review and verification. Any failure occurring during the testing shall be reported by the supplier immediately upon occurrence to the Administrative Contracting Officer's designated representative. This shall be followed, as expeditiously as possible, by the submittal in writing, of complete details of the failure, with pictures as appropriate depicting failure, supplier's proposed corrective action, extent of delay anticipated, etc., through the Administrative Contracting Officer to the Contracting Officer (PCO) for resolution and approval. Contracting Officer will determine the extent and type of retesting required to assure compliance with specification requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.1.2 Disassembly inspection. Bucket-lines of the ditchers not conforming to the performance requirements established in section 3 when tested as specified in section 4 shall be disassembled in the presence of a Government representative. Each disassembled part shall be inspected in detail for compliance with the specification and drawings in regard to materials, finishes, dimensions, and tolerances. Bucket-lines not complying with such requirements shall be rejected.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of preparation for delivery (see 4.8).

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4.3 Preproduction inspection. Prior to examination and test of the ditcher, the following operations shall be performed:

(a) In addition to the normal dynamometer test and run-in by the engine manufacturer, the engine to be installed in the pre-production model shall be adjusted and tested as follows:
  (1) Fuel supply controls shall be adjusted so that the maximum horsepower available shall not exceed the continuous horsepower rating for the applicable governed speed as established by QPL-11276 (see 3.8.1).
  (2) Fan, water pump, air compressor, and alternator as to be installed on the PPM ditcher shall be mounted on the engine.
  (3) Test the engine in accordance with test methods 2000 through 2300, MIL-STD-1400. The power run in test method 2300 shall consist of 30 minutes of stabilized speed with the dynamometer load adjusted according to the certified parasitic loads.
  (4) Following the test, the fuel control adjustment shall be sealed, and the sealed condition maintained throughout the PPM test.
(b) After installation of the engine in the ditcher, service the ditcher with military oils, greases, and fuel specified in 3.22.1 and designated for use in the ambient temperatures at which the tests will be run.
(c) Limber up the ditcher as recommended by the supplier.

4.3.1 Examination. Prior to testing, the ditcher shall be examined for the defects marked "X" in column 1 of table IV (see 4.6.1). Presence of one or more defects shall be cause for rejection.

4.3.1.1 Post-test disassembly and examination. Upon completion of all tests, the ditcher shall be disassembled and examined in accordance with 4.6.2.30. Presence of one or more defects shall be cause for rejection and corrective action (see 3.2.1).

4.3.2 Tests. Upon successful completion of the examination specified in 4.3.1, the ditcher shall be subjected to the tests marked "X" in column 2 of table V (see 4.6.2). Failure of any test shall be cause for rejection (see 3.2.1) or shall be cause for performing the inspection specified in 4.1.2, as applicable.

4.4 Initial production inspection. Unless otherwise specified (see 3.3), one or more initial production ditchers will be selected at random by the Government from the first 10 ditchers being produced from production tooling

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and will be examined for the defects marked "X" in column 2 of table IV and subjected to the tests marked "X" in column 3 of table V to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of the initial production ditcher shall not exclude the remaining ditchers from the quality conformance inspection provisions specified in section 4.

4.4.1 Inspection failure. Failure of the initial production ditcher to meet any requirements specified herein, during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production ditchers and shall be cause for refusal by the Government to continue acceptance of production ditchers until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on ditchers previously accepted and produced under the contract. Any deficiencies found as a result of the initial production testing will be considered prima facie evidence that all ditchers accepted prior to the completion of initial production testing are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Examination. After successful completion of the tests specified in 4.5.2, each ditcher shall be examined for the defects marked "X" in column 3 of table IV. Presence of one or more defects shall be cause for rejection.

4.5.2 Tests. Each ditcher shall be tested as specified in 4.6.2.31 and 4.6.2.32. Failure of either test shall be cause for rejection.

4.6 Inspection schedule.

4.6.1 Examination. The examination shall be in accordance with table IV.

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Table IV. Examination Schedule

Pre-prod.	Init. Prod.	Indiv.	Defect	Requirement Paragraph(s)
1	2	3	4	5
X	X	-	101. Dimensions or weights not as specified.	3.1.2
X	X	X	102. Assembly not complete or not correct.	Section 3
X	-	X	103. Material not as specified.	3.4
X	X	X	104. Safety devices or guards missing or not as specified.	3.6
X	X	X	105. Lubrication devices not accessible.	3.22.2
X	-	-	106. Battery enclosure not as specified.	3.20.7
X	X	-	107. Digging controls not as specified.	3.12,3.13.2, 3.13.3.2, 3.13.4.1.1
X	-	X	108. Digging boom assembly not as specified.	3.13.2
X	X	-	109. Bucket-line not as specified.	3.13.2.1
X	X	X	110. Spoil conveyor not as specified.	3.13.3 thru 3.13.3.3
X	X	-	111. Electrical system not as specified.	3.20 and table II
X	X	X	112. Lighting not as specified.	3.20.5
X	X	-	113. Cab not as specified.	3.21 thru 3.21.7
X	-	X	114. Identification or other marking or operating and safety instructions missing.	3.25, 3.26, 3.28, 3.30
X	-	X	115. Treatment and painting not as specified.	3.27
X	X	-	116. Material of incorrect thickness.	3.29
X	X	X	117. Workmanship not as specified.	3.34 thru 3.34.6

4.6.2 Tests. Tests shall be conducted in accordance with the test schedule in table V.

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Table V. Test Schedule

Item No.	Pre-prod.	Init. Prod.	Test	Test Paragraph	Requirement Paragraph
1	2	3	4	5	6
1	X	-	Overload safety release.	4.6.2.2	3.13.1.1
2	X	-	Tooth and bucket.	4.6.2.3	3.13.2.1
3	X	-	Crowd mechanism.	4.6.2.4	3.12
4	X	-	Bucket-line speed.	4.6.2.5	3.13.2
5	X	X	Conveyor belt speed.	4.6.2.6	3.13.3.2
6	X	X	Vehicle clearance circle.	4.6.2.7	3.1.2
7	X	-	Center of gravity.	4.6.2.8	3.1.2
8	X	-	Lifting attachments.	4.6.2.9	3.23
9	X	X	Tiedown attachments.	4.6.2.10	3.24
10	X	-	Towing lugs.	4.6.2.11	3.18
11a	X	-	High temperature.	4.6.2.12.1	3.5.2.1
11b	X	-	Low temperature.	4.6.2.12.2	3.5.2.1
12	X	X	Tooth replacement.	4.6.2.13	3.13.2.1
13	X	X	Service brakes.	4.6.2.14	3.19
14	X	-	Parking brake.	4.6.2.15	3.19.13
15	X	-	Rear axle capacity.	4.6.2.16	3.1.2
16	X	-	Front axle oscillation.	4.6.2.17	3.10.5
17	X	-	Fording.	4.6.2.18	3.5.5
18	X	-	Transport.	4.6.2.19	3.1.2,3.13.4.3
19	X	-	Road travel, convoy.	4.6.2.20	3.5.1,3.5.8, 3.10.1 thru 3.10.8
20	X	X	Cross-country-type travel.	4.6.2.21	3.5.1
21	X	X	Digging.	4.6.2.22	3.1.1,3.5.1, 3.12,3.13,3.14
22	X	-	Relief valve.	4.6.2.23	3.11
23	X	-	Electromagnetic interference.	4.6.2.24	3.9
24	X	-	Transportability, Berne.	4.6.2.25	3.5.7
25	X	X	Ease of maintenance.	4.6.2.26	3.7
26	X	X	Emergency steering.	4.6.2.27	3.17
27	X	-	Rain.	4.6.2.28	3.5.6
28	X	-	Storage.	4.6.2.29	3.5.2.2
29	X	-	Post-test disassembly and inspection.	4.6.2.30	N/A
30	-	X	Functional.	4.6.2.31	Section 3
31	X	-	Hydraulic system.	4.6.2.32	3.11
32	X	X	Travel speed.	4.6.2.33	3.1.2
33	X	-	Sound level.	4.6.2.34	3.8.5,3.15

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4.6.2.1 Test conditions.

(a) Test apparatus shall be accurate within the limits specified in SAE J872.
(b) Test methods referenced in tests specified herein are those contained in the appendix, part I.
(c) All lights shall be turned on and left on during all performance tests.
(d) The ditcher shall operate as specified herein without maintenance other than the supplier's recommended scheduled maintenance as established by a maintenance schedule submitted by the supplier prior to test.

4.6.2.2 Overload safety release. Test the overload safety release in accordance with test method No. 1. Nonconformance to 3.13.1.1 shall constitute failure of this test.

4.6.2.3 Tooth and bucket strength. Test the mounted teeth and tooth pockets in accordance with test method No. 2. Evidence of breakage or permanent deformation of a tooth or a tooth pocket or malfunction of the overload release mechanism shall constitute failure of this test.

4.6.2.4 Crowd mechanism (not digging). Test the crowd mechanism in accordance with test method No. 3. Nonconformance to 3.12 shall constitute failure of this test.

4.6.2.5 Bucket-line speed. Test the bucket-line speed in accordance with test method No. 4. Nonconformance to 3.13.2 shall constitute failure of this test.

4.6.2.6 Conveyor belt speed. Test the conveyor belt speed in accordance with test method No. 5. Nonconformance to 3.13.3.2 shall constitute failure of this test.

4.6.2.7 Vehicle clearance circle. Test the vehicle clearance circle in accordance with test method No. 6. Nonconformance to 3.1.2 shall constitute failure of this test.

4.6.2.8 Center of gravity. Determine the vertical and horizontal location of the center of gravity in accordance with SAE J874. Nonconformance to 3.1.2 shall constitute failure of this test.

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4.6.2.9 Lifting attachments. Test the lifting attachments in accordance with test method No. 7. Nonconformance to 3.23 shall constitute failure of this test.

4.6.2.10 Tiedown attachments, dynamic. Test the ditcher in accordance with test method No. 8. Weld failure or permanent deformation of any ditcher component or attachment or damage to any tiedown cable shall constitute failure of this test.

4.6.2.11 Towing lugs. Test the towing lugs in accordance with test method No. 9. Cracking, breaking, or permanent deformation of lug parts, connecting welds or bolts, or of the structural members to which attached shall constitute failure of this test.

4.6.2.12 Environmental operations.

4.6.2.12.1 High temperature. Test the ditcher in accordance with test method No. 10a. Inability to start within 3 minutes, inability to maintain engine temperatures within ranges specified in MIL-E-11276, failure to maintain transmission and hydraulic temperatures as specified, failure of instruments or controls, shall constitute failure of this test.

4.6.2.12.2 Low temperature. Test the ditcher in accordance with test method No. 10b. Inability to start within 3 minutes, inability to operate the engine at governed speed within 15 minutes after start, inability to maintain engine temperatures within the ranges specified in MIL-E-11276, failure of instruments, failure of controls, inability of the operator to perform the daily maintenance and daily service checks in 1 hour, or faulty operation of digging assemblies shall constitute failure of this test.

4.6.2.13 Tooth replacement. Replace all of the teeth on the bucketline utilizing two men using mechanic's common tools and special tools furnished with the ditcher. Nonconformance to 3.13.2.1 shall constitute failure of this test.

4.6.2.14 Service brakes. Test the service brakes of the ditcher in accordance with test method No. 11. Inability to make 12 consecutive stops in 50 feet or less from a speed of 25 mph shall constitute failure of this test.

4.6.2.15 Parking brake. Test the parking brake of the ditcher in accordance with test method No. 12 using either the first or alternate procedure. Nonconformance to 3.19.13 or failure to skid both rear wheels (if applicable) shall constitute failure of this test.

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4.6.2.16 Rear axle capacity. Test the rear axle capacity of the ditcher in accordance with test method No. 13. Evidence of breakage, permanent deformation, transmission oil temperature in excess of 250° F., or hydraulic oil temperature in excess of 200° F. shall constitute failure of this test.

4.6.2.17 Front axle oscillation. Oscillate the front axle and measure the amount of oscillation under the lowest point on the tire periphery. Nonconformance to 3.10.5 shall constitute failure of this test.

4.6.2.18 Fording. Test the ditcher in accordance with test method No. 14. Nonconformance to 3.5.5 shall constitute failure of this test.

4.6.2.19 Transport. Lower the boom to the surface of a carrier by means of the operating controls. Nonconformance to 3.1.2 and 3.13.4.3 shall constitute failure of this test.

4.6.2.20 Road travel, convoy. Drive the ditcher not less than 250 miles over improved highway at maximum speeds compatible with safety, including at least one 3-hour convoy run. Then drive the ditcher not less than 100 miles over unpaved roads at maximum speeds compatible with safety. Nonconformance to 3.5.1, 3.5.8 and 3.10.1 through 3.10.6 shall constitute failure of this test.

4.6.2.21 Cross-country-type travel. Test the ditcher in accordance with test method No. 15. Breakage or permanent deformation of any part shall constitute failure of this test.

4.6.2.22 Digging. Test the ditcher for digging performance in the following operations. Operate the ditcher at full capacity consistent with the type of soil and depth of excavation. Except where otherwise specified, use the follow-up scraper when digging. Replacement of the conveyor belt lacing and of worn or broken bucket teeth shall be allowed, if required, after periods of digging of not less than 50 hours for the lacing and 50 hours for the teeth. The following shall constitute failure of this test:

(a) Nonconformance to 3.1.1, 3.5.1, 3.12, 3.13 through 3.13.5, or 3.14.
(b) Evidence of any breakage, malfunction, or permanent deformation of the engine, transmission system, drive trains, running gear, overload safety release, digging-boom assembly, spoil conveyor, hydraulic components, controls, or structural components.

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(c) Loss of more than two teeth during any 50 consecutive hours of digging.
(d) Failure of the conveyor belt lacing before 50 hours of digging.
(e) Teeth worn beyond use before 50 hours of digging.

In case of failure of the digging test, the supplier shall remove the deficient component or assembly, determine the cause of the failure, and correct the deficiency. During this deficiency correction, the supplier will be permitted to replace any other component or assembly on the ditcher. Following correction of the deficiency, perform the complete digging test without considering the time or type of digging accomplished prior to failure.

4.6.2.22.1 Sidehill slopes. Test the ditcher in accordance with test method No. 16.

4.6.2.22.2 Uphill slopes. Test the ditcher in accordance with test method No. 17.

4.6.2.22.3 Downhill slopes. Test the ditcher in accordance with test method No. 18.

4.6.2.22.4 Curved trench. Test the ditcher in accordance with test method No. 19.

4.6.2.22.5 Foxholes. Test the ditcher in accordance with test method No. 20.

4.6.2.22.6 Ditches. Test the ditcher in accordance with test method No. 21.

4.6.2.23 Relief valve(s). Test the setting of the hydraulic system relief valve or valves in accordance with test method No. 22. Nonconformance to 3.11 shall constitute failure of this test. At the supplier's option, this test may be performed by conducting a bench test or tests prior to assembly of the ditcher.

4.6.2.24 Electromagnetic interference. Test the preproduction model in accordance with SAE J551. The supplier shall furnish to the contracting officer a copy of the test report. Disapproval of the report shall constitute failure of this test.

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4.6.2.25 Transportability, Berne. Sectionalize the ditcher. Cutting and welding are not permitted. Nonconformance to 3.5.7 shall constitute failure of this test. This test may be conducted in conjunction with 4.6.2.19.

4.6.2.26 Ease of maintenance. Test the ditcher in accordance with test method No. 23. Inability of one man to complete any one operation on the maintenance-operations list in the time allotted or nonconformance to 3.7 shall constitute failure of this test.

4.6.2.27 Emergency steering. Test the ditcher in accordance with test method No. 24. Inability of the ditcher to negotiate the test course or failure to make contact with all markers on the test course shall constitute failure of this test.

4.6.2.28 Rain. Test the ditcher in accordance with test method No. 25. Any of the following shall constitute failure of this test:

(a) Water in the air intake filters and bowl in excess of 8 percent of the fluid capacity of the sediment chamber.
(b) Water leakage into the torque converter oil chambers, engine crankcase sump, engine exhaust manifold or turbo exhaust port (when applicable), transmission, transfer case, instruments, gages, or fuel tank.
(c) Malfunctioning of any electrical component.
(d) Inability to start the engine within 2 minutes.
(e) Malfunctioning of any gage or instrument.

4.6.2.29 Storage. Test the ditcher in accordance with test method No. 26. Nonconformance to 3.5.2.2 shall constitute failure of this test. Certification by the supplier of satisfactory performance of the ditcher after storage in ambient temperatures specified herein in lieu of test may be acceptable when approved by the contracting officer.

4.6.2.30 Post-test disassembly and examination. Upon completion of all tests, measure and record the compression in each engine cylinder, and dismantle the ditcher sufficiently to permit detailed disassembly and visual examination of the following components:

(a) Engine - Cylinder head(s), all valves, two pistons and piston pins from the two cylinders having the lowest compression readings at the end of the test, one main bearing and one connecting rod bearing, the turbocharger (if used), and the oil in the crankcase.

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(b) Torque converter and transmissions.
(c) Universal joints and drive shafts.
(d) Bucket-line.
(e) Final drive planetary, one front, one rear.
(f) Complete brake system.
(g) Steering system cylinders, valves, and pump.
(h) Hydraulic system - pumps, motors, valves, one cylinder of each size, control levers, and any hose and fitting where leakage occurred during the tests.
(i) One wheel and wheel bearing.
(j) Digging boom assembly.
(k) Spoil conveyor assembly.
(l) Tire, wheel assembly, one front and one rear.
(m) Overload safety release mechanism and bucket-line drive train.

Record a complete description of the condition of all parts and assemblies examined, and take pictures of any deficiency. Evidence of corrosion which affects the functions of the ditcher, evidence of permanent deformation, lack of lubrication where required, breakage, malfunction, or excessive wear of any component shall constitute failure of this test. Manufacturing tolerances listed in the supplier's quality control program shall be used as the basis for determining excessive wear.

4.6.2.31 Functional. Test each ditcher in accordance with test method No. 27. Nonconformance to 3.1.2, 3.5.1, 3.11, 3.12, 3.13.1, 3.13.1.1, 3.13.2, 3.13.3, 3.19, or 3.19.3 shall constitute failure of this test.

4.6.2.32 Hydraulic system. Test the hydraulic system for cleanliness and leaks in accordance with test method No. 28. Evidence of metallic slag or other contamination harmful to the system; malfunction of any pressure relief valve; or evidence of leakage that cannot be stopped as described in 2(e) of the test method, shall constitute failure of this test.

4.6.2.33 Travel speed. Travel speeds for low and high forward gear and reverse gear shall be determined in accordance with test method No. 29. Nonconformance of travel speeds to requirements of table I shall constitute failure.

4.6.2.34 Sound level. Test the ditcher in accordance with test method No. 30 to determine the sound levels when stationary and moving. The test shall be conducted in accordance with SAE J919. Nonconformance to 3.15 shall constitute failure of this test.

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4.6.2.35 Hydraulic components (when required, see 3.2.2).

4.6.2.35.1 Hydraulic motor. The hydraulic motor(s) shall be tested in accordance with test method No. 31. Failure to operate continuously, inability to meet proof pressure, efficiency less than specified, external leakage of oil, galling, scoring, or inability to complete the test shall constitute failure of this test.

4.6.2.35.2 Hydraulic suction line. The reservoir-to-pump suction line assembly shall be tested in accordance with test method No. 31. A change in length of the hose in excess of that specified in 3.11.5, evidence of defects or deterioration of the hose, evidence of leakage, or evidence of slippage of the hose on the tube shall constitute failure of this test.

4.6.2.35.3 Hydraulic pump. Test the hydraulic pump as specified in MIL-P-52675, including failure criteria therein.

4.7 Inspection comparison. The Government may select ditchers at any time during the contract production period and subject these ditchers to the examination and tests specified in 4.4 to determine that the quality of the selected ditchers is equal to the quality standards established during initial production testing. The inspection will be performed by the Government at a site selected by the Government. Ditchers will be selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production ditcher.

4.7.1 Inspection failure. Failure of any ditcher when inspected as specified in 4.7 may be considered cause for refusal by the Government to continue acceptance of ditchers until objective evidence furnished by the supplier reveals that corrective action has been taken to eliminate the condition which caused the rejection.

4.8 Inspection of preparation for delivery.

4.8.1 Preproduction pack inspection.

4.8.1.1 Examination. Examine the preproduction pack for the defects specified in 4.8.2.3. Presence of one or more defects shall be cause for rejection.

4.8.1.2 Test. The preproduction pack of crated components for level A shall be subjected to the railroad-car test as specified for guided-impact tests in accordance with MIL-STD-1186, Appendix A. The car shall strike

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a string of five empty cars with draft gear extended and the brakes set, at a speed of not less than 10 miles or more than 11 miles per hour. At the conclusion of the preproduction pack test, the pack shall be examined. Any shifting of contents, loosening or breaking of holddowns, ties, stays, blocking or bracing, or any visual damage to the components shall constitute failure of the preproduction pack and shall be cause for rejection.

4.8.2 Quality conformance inspection of pack.

4.8.2.1 Unit of product. For the purpose of inspection, a completely processed ditching machine prepared for shipment shall be considered a unit of product.

4.8.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.8.2.3 Examination. Samples selected in accordance with 4.8.2.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

118.	Materials, methods, and containers not as specified for level A or B. Each incorrect material, method, or container shall be considered one defect.
119.	Disassembly not as specified.
120.	Removed bolts, nuts, pins, and washers not replaced in one of the mating parts and secured to prevent loss as specified.
121.	Removed keys not secured as specified.
122.	Cloth matchmarking tags not waterproofed as specified for level A.
123.	Surfaces of components requiring a contact preservative in accordance with the referenced document not coated for level A or B.
124.	Machined surfaces not preserved and wrapped or covered as specified for level A or B.
125.	Carrier section not preserved and packaged in accordance with the referenced document and additional requirements specified.
126.	Gear housings not filled with lubricant and the gears not actuated as specified for level A or B.
127.	Drive chains not preserved as specified for level A or B.
128.	Hydraulic system not preserved as specified for level A or B.
129.	Conveyor belt not removed or released from tension for level A or B.
130.	Removed conveyor belt not secured as specified for level A or B.
131.	Maintenance tools not preserved and packaged in accordance with the referenced document for level A or B.

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132. Consolidated packaging not as specified for level A or B.
133. Contents not blocked and braced within the boxes to prevent free movement for level A or B.
134. Toolbox lid not secured to prevent pilferage.
135. Packing not as specified for level A.
136. Strapping not zinc coated for level A.
137. Depreservation guides not prepared and attached as specified.
138. Marking missing, illegible, incorrect, or incomplete.

5. PREPARATION FOR DELIVERY

5.1 Preproduction pack. The supplier shall furnish a preproduction pack for examination and test within the time frame specified to prove, prior to starting production packaging that the applied preservation, packaging, packing and marking comply with the preparation for delivery requirements of this specification. Examination and test shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.6). The preproduction pack may be accomplished utilizing either the preproduction model ditcher or a production ditcher. If the preproduction model ditcher is utilized, any preservation, packaging, and packing shall be removed by the supplier at no expense to the Government, when requested by the Government, to facilitate comparison between the preproduction ditcher and production ditchers.

5.2 Preservation and packaging. Preservation and packaging shall be level A, B, or C as specified.

5.2.1 Level A.

5.2.1.1 Disassembly and matchmarking.

5.2.1.1.1 Disassembly. The ditcher shall be disassembled as required for transport; further disassembly shall be the minimum necessary to safeguard parts subject to damage or loss and to accomplish reduction in cube, in accordance with the supplier's recommendations. Removed bolts, nuts, pins, and washers shall be replaced in one of the mating parts and secured to prevent their loss. Keys removed shall be secured to the keyway of the primary component or shaft.

5.2.1.1.2 Matchmarking. When necessary to facilitate reassembly, detached components shall have matchmarking information on cloth tags conforming to UU-T-81, type A, or on metal tags. Cloth tags shall be waterproofed in accordance with MIL-STD-129. The tags shall be securely attached to the components so as to prevent damage to the preservation or the components.

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5.2.1.2 Preservatives. Preservatives specified herein shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116.

5.2.1.3 Unprotected metal surfaces.

5.2.1.3.1 Unfinished surfaces. Unpainted exterior unfinished metal surfaces of items such as pins, linkages, universal joints, sprockets, shafts, buckets, wiper plates, cutting teeth, and any other unpainted exterior metal surfaces not specified herein requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative.

5.2.1.3.2 Machined surfaces. Exposed exterior machined metal surfaces shall be coated with type P-6 preservative. The coated surfaces shall be wrapped or covered with barrier material conforming to MIL-B-121, grade A, type I, class 2, and secured in place with tape conforming to PPP-T-60, type IV.

5.2.1.4 Carrier section. Components of the carrier section shall be preserved and packaged in accordance with the level A (mobile) requirements of MIL-STD-281, and the following.

5.2.1.4.1 Batteries. Batteries shall be secured in place in the battery carrier.

5.2.1.4.2 Cooling system. The cooling system shall be preserved by the water and antifreeze procedure.

5.2.1.5 Digger section.

5.2.1.5.1 Enclosed gears. Gear housings shall be filled to the operating level with the operational lubricant and the gears shall be actuated to assure coating all internal parts and surfaces with the lubricant.

5.2.1.5.2 Drive chains.

5.2.1.5.2.1 Exposed chains. Exposed drive chains shall be coated with enough type P-3 or type P-9 preservative to assure penetration to the inner surfaces of the roller, pins, and bushings. After the excess preservative has drained, the chain shall be coated with type P-1 preservative.

5.2.1.5.2.2 Enclosed chains. Enclosed drive chains shall be preserved as specified in 5.2.1.5.1 for enclosed gears.

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5.2.1.5.3 Hydraulic system. The hydraulic fluid supply reservoir shall be filled to the operating level with the operational fluid. The pistons shall be retracted into the cylinders and secured. Any exposed portions of the piston rods shall be coated with type P-6 preservative and wrapped or covered with barrier material conforming to MIL-B-121, type I, grade A, class 2, extending the wrap approximately 2 inches on the ram cylinder. The wrap shall be secured in place with tape conforming to PPP-T-60, type IV. The hydraulic control valves shall be secured in the neutral position and preserved as specified for the piston rods. Hydraulic lines shall not be disconnected unless required to meet the transportability requirements. If disconnected, openings shall be sealed with plastic caps or plugs as applicable. Any other exterior metal surface of components of the system requiring the application of a contact preservative shall be coated with type P-1 preservative.

5.2.1.5.4 Conveyor belt. The conveyor belt, if not removed to comply with the disassembly requirements shall be released from tension. Removed belts shall be compactly rolled, secured with strapping conforming to QQ-S-781, and consolidated packaged as specified in 5.2.1.8.

5.2.1.6 Repair parts. The preservative application criteria and applicable method(s) of preservation of MIL-P-116 shall be used to preserve repair parts.

5.2.1.7 Maintenance tools. Maintenance tools shall be preserved and packaged in accordance with the level A requirements of MIL-H-15424 and placed in the toolbox, if space permits. When sufficient space is not available, the tools shall be consolidated packaged as specified in 5.2.1.8.

5.2.1.8 Consolidated packaging. Packaged and unpackaged components removed by disassembly requiring the protection of a box, and the repair parts and maintenance tools that will not fit in the toolbox shall be intermediate packaged in close-fitting boxes conforming to PPP-B-621, class 2, style optional, or PPP-B-601, overseas type, style optional. The contents shall be blocked and braced within the boxes to prevent free movement and damage to the contents. The number, size and weight of the boxes shall be determined by the available space and convenience for packing or securing the boxes on the ditcher. Strapping of the boxes shall not be required.

5.2.2 Level B. Each complete ditcher shall be preserved and packaged as specified for level A with the following exceptions.

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5.2.2.1 Matchmarking. Cloth matchmarking tags shall not be required to be waterproofed.

5.2.2.2 Carrier section. Components of the carrier section shall be preserved and packaged in accordance with the level B (mobile) requirements of MIL-STD-281, and with the additional requirements specified for level A.

5.2.2.3 Exposed chains. Exposed chains shall be preserved as specified for level A, except only type P-3 or type P-9 preservative shall be applied.

5.2.2.4 Consolidated packaging. Boxes shall be domestic type or class as applicable.

5.2.3 Level C.

5.2.3.1 Disassembly. Disassembly shall be as specified in 5.2.1.1.1.

5.2.3.2 Carrier section. Components of the carrier section shall be preserved and packaged in accordance with the level C (mobile) requirements of MIL-STD-281, and with the additional requirements specified for level A.

5.2.3.3 Repair parts and maintenance tools. Repair parts shall be preserved and packaged as specified for level A. Maintenance tools shall not require preservation and packaging but shall be placed in the toolbox, and the toolbox lid shall be secured to prevent pilferage.

5.2.3.4 Other components. Preservation and packaging of other components of the ditcher not specifically mentioned shall be limited to that protection deemed necessary by the supplier to provide protection against deterioration and damage from the supplier to the initial destination.

5.3 Packing. Packing shall be level A or C as specified.

5.3.1 Level A. When the ditcher section is detached from the carrier section to meet the transportability requirements, the carrier section of the ditcher shall be prepared for delivery mobile. The ditcher section, consolidated containers, and unpackaged components shall be packed in a crate conforming to MIL-C-104, type I, class 1 or 2, style a or b. Blocking, bracing, and anchoring of crated components shall be in accordance with MIL-STD-1186 and with the appendix to the crate specification. When the ditcher section does not have to be detached from the carrier section, the complete

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ditcher shall be prepared for delivery mobile. Unpackaged loose components and consolidated containers shall be secured to the ditcher with strapping conforming to QQ-S-781, type I, class B, in a manner that does not increase cubage or interfere with the mobility of the ditcher. Toolbox lids shall be secured in a manner to prevent pilferage.

5.3.2 Level C. Each complete ditcher with repair parts and tools shall be prepared for delivery in a manner to assure carrier acceptance and safe delivery to destination at lowest rating in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.4 Depreservation guide. Two depreservation guides shall be prepared. One of the guides shall be preserved in accordance with method IC-1. The bag shall be marked "Depreservation Guide" and secured to the ditcher in a conspicuous location. The other copy shall be prepared and placed in the package with the technical publications. Unless otherwise specified, DA Form 2258 - Depreservation Guide for Vehicles and Equipment shall be used (see 6.7).

5.5 Marking. Packages, shipping containers, and mobile ditchers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The ditcher is intended for use in general-purpose ditching in the type of soil specified herein, and digging foxhole and slit trenches.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of preproduction model (see 3.2.1).
(c) When initial production inspection is required and number of ditchers to be furnished, when applicable (see 3.3).
(d) Applicable list of repair parts and maintenance tools (see 3.31).
(e) When certification in lieu of the storage test is acceptable (see 4.6.2.29).
(f) Time frame required for submission of preproduction pack (see 5.1).

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(g) Level of preservation and packaging and level of packing required (see 5.2 and 5.3). Level B preservation and packaging is intended to provide economical but limited protection and should be specified only when it is determined the ditchers will be held in covered storage.
(h) When other than DA Form 2258 is to be used (see 5.4).

6.3 Government-furnished property. The contracting officer should arrange to furnish to the supplier the property specified in 3.32.

6.4 Technical publications. Technical publications should contain the following information:

(a) Special steels. When high-strength alloy steels are used, such alloys should be identified in the technical publications. Welding repair procedures for these alloy steels should be included in the technical publications.
(b) Storage. Instructions to prevent damage occasioned by prolonged storage should be included in the technical publications.

6.5 Preproduction model. Any changes or deviations of production ditchers from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish ditchers conforming to this specification.

6.6 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the supplier of his obligation to preserve, package, pack, and mark the ditchers in accordance with this specification.

6.7 DA Form 2258. The contracting officer should arrange to furnish DA Form 2258 when requested by the supplier.

6.8 Data questionnaire. The contracting officer should require the supplier to complete and submit the questionnaire, part II of the appendix to this specification, prior to preproduction model testing as evidence of compliance with the requirements.

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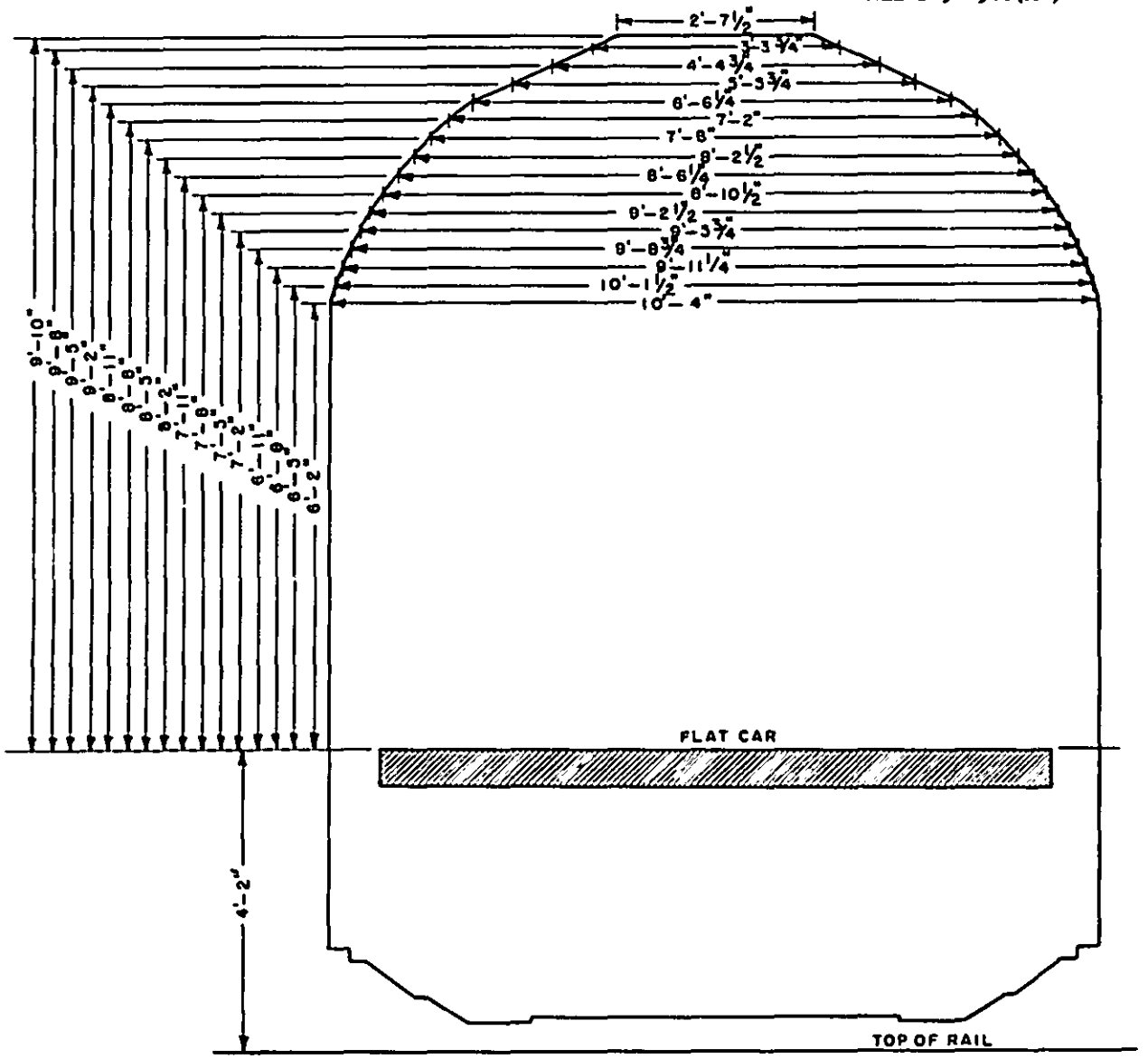
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Preparing activity:

Army - ME

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ITEM LENGTH-32'-10"

FIGURE 1. BERNE INTERNATIONAL
(CLEARANCE DIAGRAM)

A 13123-1

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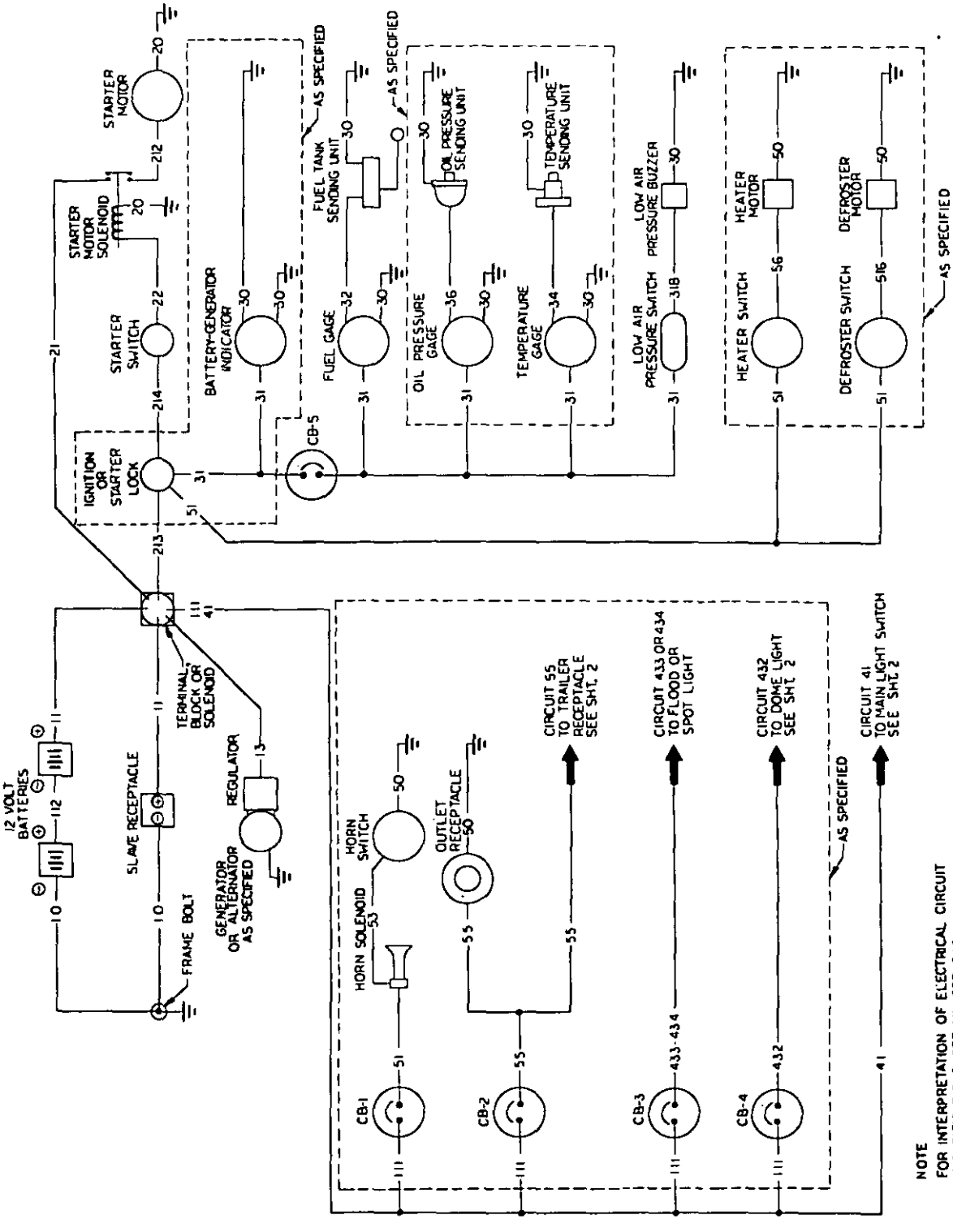


FIGURE 2. SCHEMATIC WIRING DIAGRAM, POWER AND MONITOR X-184

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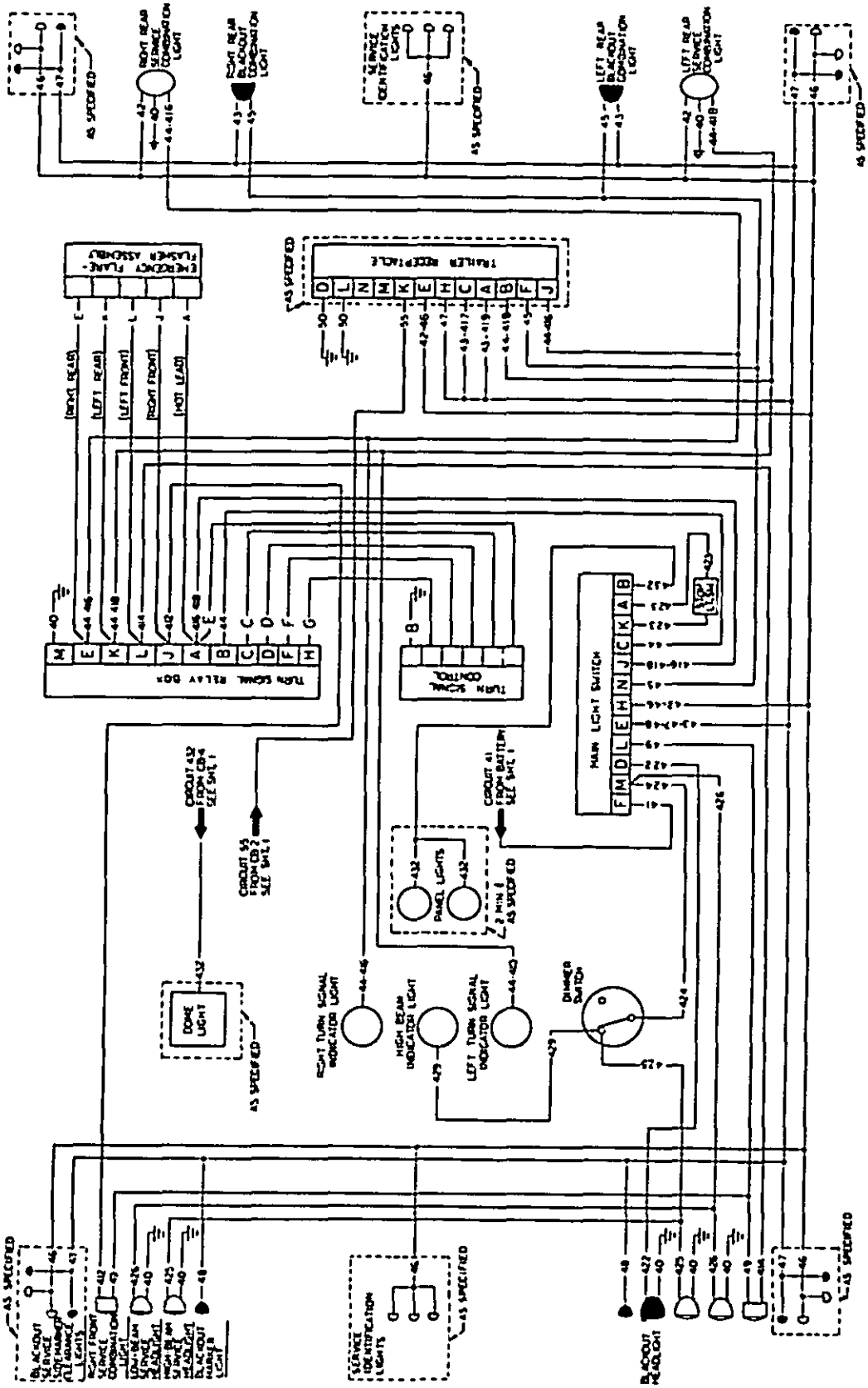


FIGURE 3. SCHEMATIC WIRING DIAGRAM, LIGHTING

X-185

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APPENDIX

FOR

DITCHING MACHINE: DIESEL-ENGINE-DRIVEN: WHEEL-MOUNTED; LADDER

PART I TEST METHODS

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TEST METHOD NO. 1

OVERLOAD SAFETY RELEASE

1. TEST EQUIPMENT:

Heavy chain or cable with hook; load cell, 20,000-pound capacity and recording instrumentation; deadman.

2. TEST PROCEDURE:

(a) Hook the chain or cable to a bucket on the digging side of the ladder. Connect the load cell to the chain or cable, and anchor it to the deadman. Arrange the position of the set-up to measure the pull in the direction parallel to bucket travel.

(b) At engine idle, shift the transmission into a digging speed gear. Gradually accelerate the engine while recording the pull on the bucket. When the pull reaches or slightly exceeds 12,000 pounds, hold the engine speed constant for 10 seconds, then continue to accelerate until the release mechanism begins to operate. Return engine to idle speed.

(c) Repeat (b) above a second and third time.

(d) Record pull at initial release, pull while slipping, and duration of release operation.

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TEST METHOD NO. 2

TOOTH AND BUCKET STRENGTH

1. TEST EQUIPMENT:

A restraining snag, such as a rail, beam or tooth socket device, solidly anchored. Its shape and location shall permit test personnel to hook any single tooth of any bucket onto the snag when the bucket being tested is between the bottom of the boom and a point 5 feet up along the digging path. The single restrained tooth shall be the only resistance to bucket-line movement. The engagement of the tooth in the restraint shall not exceed 3/4 inch from the tooth tip.

2. TEST PROCEDURE:

(a) Check the ditching machine for correct overload safety release setting as specified in 4.6.2.2.

(b) Shift the boom into digging position, move the ditcher into position, and hook one bucket tooth onto the restraining snag described above.

(c) Shift the transmission to provide the highest bucket-line speed.

(d) Accelerate the engine slightly to provide enough torque to remove all slack from the drive train, bucket-line, and tooth restraint. Then accelerate the engine until the overload release clutch slips. Maintain that condition for 2 to 3 seconds before reducing throttle to idle.

(e) Repeat (b), (c), and (d) above on succeeding buckets with the restraining force applied to a tooth in a different position on the bucket each time. Continue until a tooth in each position in the digging pattern has been tested.

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TEST METHOD NO. 3

CROWD MECHANISM

(NOT DIGGING)

1. TEST EQUIPMENT:

   (a) Stopwatch for timing speed of travel.

   (b) Stakes or other means for marking distances along the courses.

   (c) Measuring tape.

2. TEST COURSES:

   (a) Straight, level track not less than 50 feet long, paved or unpaved.

   (b) Natural slope or prepared ramp having a grade of not less than 27 percent. The length of the slope at the required grade shall be not less than 50 feet.

3. TEST PROCEDURE:

   (a) Stake out, or otherwise mark, the distances along both courses at 10-foot intervals.

   (b) With the digging boom in digging position and raised to provide ground clearance, place the ditcher in crowd and travel at maximum crowd speed for a distance of not less than 50 feet. Record the maximum crowd speed.

   (c) Adjust the crowd speed in low range to travel at a speed of not less than 14 feet per minute. Record this speed. While traveling at this control valve setting, operate the conveyor belt and conveyor shift continuously throughout the entire range of each component. Record this speed.

   (d) With the ditcher facing downgrade, use the forward crowd to move the machine down the 27 percent grade at speeds of 10, 15, and 20 feet per minute measured over the first 10 feet. Without changing the speed control, measure and record the ditcher speed over the first and the fourth 10-foot sections of the course.

   (e) Repeat (d) above, except with the ditcher facing upgrade and using reverse crowd.

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TEST METHOD NO. 4

BUCKET-LINE SPEED

1. TEST EQUIPMENT:

Stopwatch; measuring tape, or other means to determine accurately the length of the bucket-line; paint and brush to paint bucket and boom.

2. TEST PROCEDURE:

(a) Paint one bucket and an index line on the boom.

(b) Determine length of bucket-line.

(c) Place the transmission in gear to drive the bucket-line at the lowest speed in the digging direction. Accelerate the engine to full throttle and operate at governed speed.

(d) When the bucket-line reaches constant speed, measure the time required for 5 circuits of the bucket-line. Calculate bucket-line speed.

(e) Repeat (b) and (c) above, except with the transmission shifted to successively higher gears including the highest speed.

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TEST METHOD NO. 5

CONVEYOR BELT SPEED

1. TEST EQUIPMENT:

Stopwatch; measuring tape, or other means to determine accurately the length of the conveyor belt; paint and brush to mark section of belt.

2. TEST PROCEDURE:

(a) Determine length of conveyor belt.

(b) Paint one section of the belt between two adjacent cleats.

(c) Run the engine at governed speed.

(d) With the discharge direction control in "left discharge" position, gradually increase the conveyor belt speed from zero to top speed in 1 minute, then gradually reduce speed to zero in 1 minute.

(e) Repeat (d) except with the discharge direction control in "right discharge" position.

(f) Operate the conveyor belt at top speed. Measure the time required for 20 circuits. Calculate conveyor belt speed.

(g) Repeat (f) except with the discharge direction control in the "left discharge" position.

(h) With the conveyor belt running at top speed, quickly reverse the discharge direction control, causing the conveyor belt to stop and resume travel immediately in the opposite direction. Repeat this reversal of discharge direction control every 15 seconds for 2 minutes.

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TEST METHOD NO. 6

VEHICLE CLEARANCE CIRCLE

1. TEST EQUIPMENT:

(a) 100-foot measuring tape.

(b) Plumb line or rod attached to the ditcher in vertical position.

(c) 6-foot straight edge.

(d) Markers and chalk.

2. TEST COURSE:

The course shall be dry, level concrete or firm ground large enough to accommodate the ditcher when negotiating a turn of not less than 270 degrees (see figure A-1).

3. TEST PROCEDURES:

(a) Adjust tire pressures to recommended operating pressure.

(b) Drive the ditcher slowly in a sharp curve and determine the part of the ditcher which extends farthest from the center of the turning circle. Attach the plumb bob or vertical rod at that point.

(c) Operate the ditcher to make a turn to the right of not less than 270 degrees while traveling forward in low gear with the engine at idle speed.

(d) During step (c) mark on the course the path described by that part of the ditcher which extends farthest from the center of the circle.

(e) Measure the diameter of the marked path in three directions not less than 20 degrees apart. Take the average of the three measurements, record this average measurement as D-c1 (see figure A-1).

(f) Repeat (c), (d), and (e) above, except the 270-degree turn shall be made to the left.

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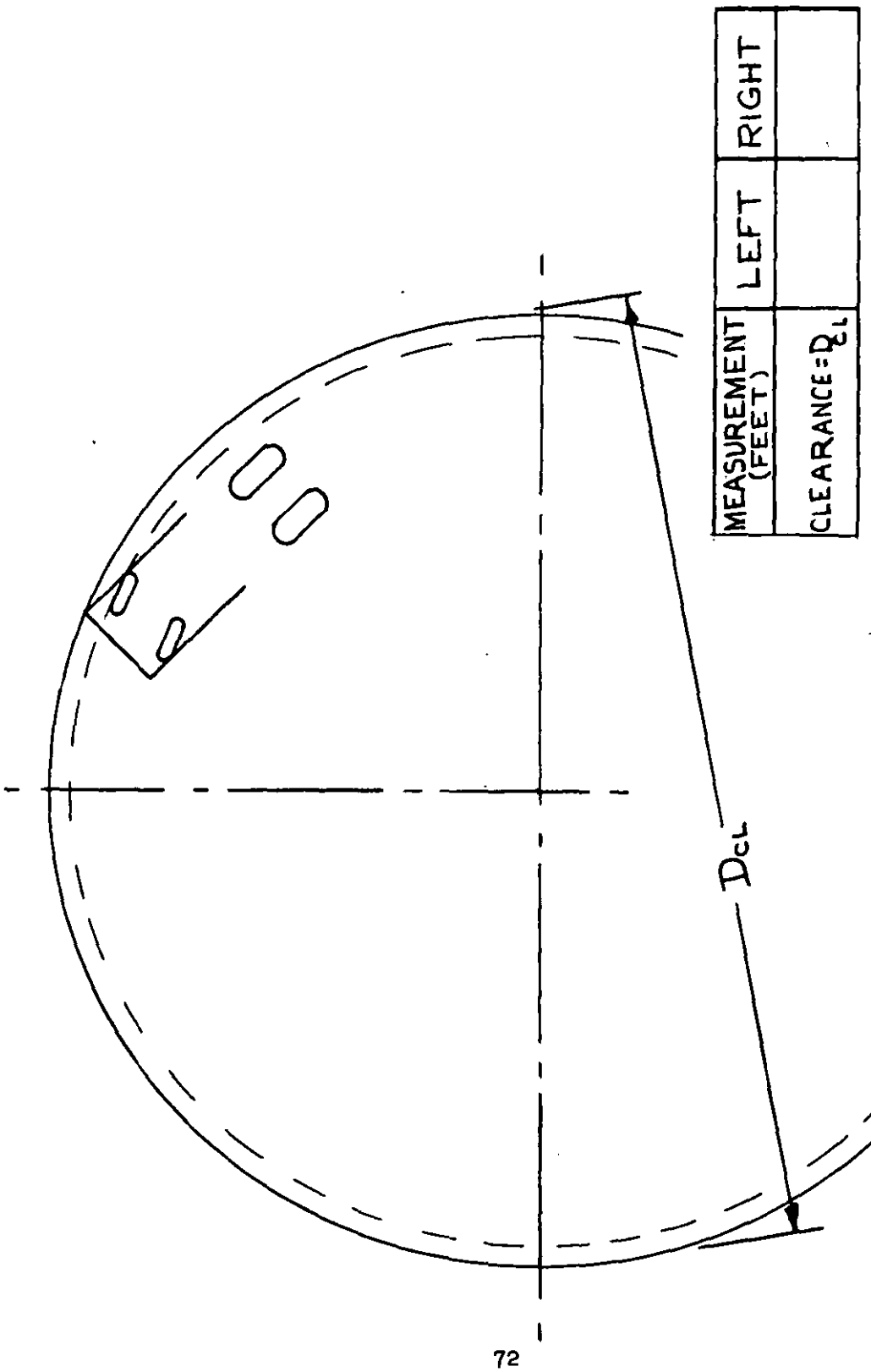


FIG. A-1 COURSE FOR TURNING CLEARANCE

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TEST METHOD NO. 7

LIFTING ATTACHMENT

1. TEST APPARATUS:

(a) Crane or lifting device.

(b) Slings with hooks.

2. TEST PROCEDURE:

(a) Oscillation stop blocks may be affixed to the ditcher to limit oscillation of the front axle.

(b) Lift the ditcher and hold in suspension for at least 3 minutes in its normal travel position using slings that converge not more than 19 feet above the lowest extremity of the ditcher. Do not use spreader bars. Determine the force and the angles of application for each attachment. Measure the clearance between each sling and the ditcher.

(c) Restrain the ditcher by anchoring the main frame or axles and subject each attachment to a load applied in the direction as determined in (b) above and equal to 2-1/2 times the force determined in (b) above. Hold each load a minimum of 90 seconds.

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TEST METHOD NO. 8

TIEDOWN ATTACHMENTS (DYNAMIC)

1. TEST APPARATUS:

   (a) Timing devices.

   (b) Tiedown material - cable not less than 1/2-inch size.

   (c) Six railroad flatcars.

2. TEST PROCEDURE:

   (a) Anchor the ditcher, positioned for rail shipment, on a flatcar using only the tiedown provisions. Use no blocking except for vertical support under the digging boom. Hump the car, with the ditcher tied down, at a speed of not less than 7 mph against five cars with their brakes set by air. The ditcher shall face in the direction of the five cars.

   (b) Repeat the procedure with the rear of the ditcher in the direction of the five cars.

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TEST METHOD NO. 9

TOWING LUGS

1. TEST EQUIPMENT:

   (a) Towbar and clevises as specified in 3.18, or a suitable clevis and cable to apply load to a single lug.

   (b) Load cell, or other means of determining the loads applied to the lugs.

   (c) Means for applying and holding a constant load.

2. TEST PROCEDURE:

Apply the static load developed by a 35-ton pull on the towbar and clevises specified in 3.18 to each mounted towing lug. If the load is applied by a single line in place of the towbar, exert a forward pull of 36,000 pounds in the horizontal plane. Exert the pull through a suitable clevis attached to the towing lug. Apply this load to each lug.

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TEST METHOD NO. 10a

HIGH TEMPERATURE

1. TEST EQUIPMENT:

(a) Test chamber to maintain specified temperature, with exhaust system to remove engine exhaust fumes.

(b) Thermocouples, recorders for the thermocouples.

(c) Blocking for front and rear wheels.

2. TEST PROCEDURE:

Service the ditcher with military lubricants and fuel suitable for high-temperature operation (see 4.6.2.12.1) and fill the cooling system with water. Install thermocouples to measure the temperatures specified below. Secure the bucket-line from moving by fastening a heavy chain around it as in test method No. 1. Set the parking brake. Block the wheels. Subject the ditching machine to an ambient temperature of not less than 120° F. until all parts are stabilized at that temperature. Start the engine and run without load for not more than 15 minutes. After the no-load warmup, operate the engine at governed speed for 2 hours, and load the engine during the 2-hour operation as follows:

(a) Drop engine speed to low idle. Shift the transmission to provide highest bucket-line speed. Gradually accelerate the engine until the release mechanism begins to slip. Immediately reduce engine speed to low idle. Shift transmission to neutral. Perform the above operating cycle four times during the 2-hour test.

(b) Apply the service brakes. Operate the crowd control in forward and reverse. Increase the crowding force gradually until the wheels begin to move, overriding the brakes, or until the pressure-relief valve is actuated, whichever occurs first. Operate the crowd mechanism in this manner once every 5 minutes during the 2-hour test, alternating between forward and reverse crowd. During the warmup and 2-hour-operation periods, record continuously, or at intervals not greater than 10 minutes, the following temperatures: (1) Engine coolant in engine coolant jacket discharge pipe and coolant inlet to the engine; (2) engine lubricating oil in the oil gallery; (3) hydraulic oil in the reservoir; (4) transmission oil; (5) electrolyte in a central cell in each battery; and (6) ambient temperatures not more than 3 feet from each corner of the ditcher at a height above ground between 2 and 3 feet.

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TEST METHOD NO. 10b

LOW TEMPERATURE

1. TEST EQUIPMENT:

(a) Low-temperature test chamber.

(b) Thermocouples and recorder for thermocouples.

(c) Arctic clothing.

2. TEST PROCEDURE:

Service the ditcher with arctic grade military lubricants and fuel conforming to VV-F-800, and fill the cooling system with antifreeze mixture. Block up the ditcher to raise the rear wheels clear of the floor. Subject the ditcher to a temperature of minus 25° F. or lower until all parts are stabilized at that temperature. Have the operator, clothed in winter clothing and wearing mittens, perform the daily maintenance and daily service checks. Crank the engine using the batteries mounted on the ditcher without outside power source. Run the engine without load for not more than 15 minutes from time of start. Maintain the ambient temperatures at the locations specified below at minus 25° F. throughout the test, except that locations adjacent to the engine compartment may rise. After the no-load warmup, turn on the cab heater and operate the engine as follows:

(a) 10 minutes at half throttle.

(b) 5 minutes at governed speed.

(c) 10 minutes at idle.

Perform this cycle three times. During the cycling operation at half throttle and full throttle, operate all controls and moving parts. Operate the bucket-line at the lowest speed range for two continuous runs of 15 minutes each. Operate the conveyor belt at half speed for two continuous runs of 15 minutes each. Operate the crowd control to cause not less than one complete turn of both rear wheels in each direction. During warmup period and the 1-hour operation, record continuously, or at intervals not greater than 10 minutes, the following temperatures:

(a) Engine coolant in engine-coolant-jacket discharge pipe and coolant inlet to the engine.

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(b) Engine lubricating oil in the oil gallery.

(c) Hydraulic oil in the reservoir.

(d) Electrolyte in a central cell in each battery.

(e) Ambient temperatures not more than 3 feet from each corner of the ditcher at a height above ground between 2 and 3 feet.

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TEST METHOD NO. 11

SERVICE BRAKE

1. TEST COURSE:

The test course shall be a uniform improved surface allowing no penetration, with a grade not exceeding plus or minus 1 percent longitudinally and shall be sufficiently long to permit the ditcher to attain a speed of 25 mph.

2. TEST APPARATUS:

(a) Tape, measuring (100-foot).

(b) Chalk.

(c) Detonator, 2-barrel, brake apparatus.

(d) Blanks, .22 cal., yellow powder.

(e) Towed fifth wheel.

3. TEST PROCEDURE:

(a) Connect the detonator as follows:

(1) First barrel - actuated by a switch directly connected to the brake pedal so that it is actuated during the first 1/4 inch of pedal travel.

(2) Second barrel - actuated by a switch operated manually after the ditcher has come to a complete stop.

(b) Drive the ditcher with driver and assistant over the test course. When the ditcher reaches a speed of 25.0 mph, the assistant shall signal the driver to apply the brakes. Measure the distance between the point of brake pedal application and point of stoppage. Perform the test 12 times, six times in each direction, allowing not more than 3 minutes between each braking stop. All tests shall be made from a speed of not less than 25 mph.

NOTE: A brake tester apparatus approved by the Government may be used in place of the detonator and switches.

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TEST METHOD NO. 12

PARKING BRAKE

1. TEST COURSE:

The test course shall be an improved surface area with a minimum longitudinal grade of 35 percent with a minimum length equal to three times the overall ditcher length.

2. TEST PROCEDURE:

(a) Drive the ditcher onto the test course with the front of the ditcher in the downhill direction and stop. Apply the parking brake and dismount.

(b) Repeat the test with the ditcher positioned with the front in the uphill direction.

3. ALTERNATE PROCEDURE:

(a) Using a towing vehicle, load cell, and cable connected to the towing lugs, determine breakaway force, "F", required to move ditcher with the parking brake set, when positioned on a level surface with the gearshift in neutral and the service brake is not applied.

(b) Determine the equivalent slope by the formulas:

(1) $F = W \sin a$

where F = breakaway force
W = weight of ditcher
a = slope angle in degrees

(2) Slope, percent = 100 X tan a

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TEST METHOD NO. 13

REAR AXLE CAPACITY

1. TEST EQUIPMENT:

(a) Weights and mounting attachments for fastening weights to rear axle and frame.

(b) Heavy truck scale to weigh rear axle load.

(c) Thermocouples and portable potentiometer.

2. TEST COURSE:

Cross-country-type travel course - see test method No. 15.

3. TEST PROCEDURE:

Load the rear axle to not less than 38,000 pounds with the machine in travel position. Install thermocouples in the lubricating oil sump of the secondary transmission and in the hydraulic reservoir, and record the temperature continuously or at intervals not greater than 10 minutes during this test. Place the ditcher in crowd drive. Operate the ditcher at maximum crowd speed over the same course used for the cross-country road test (see 4.6.2.21) for a period of not less than 3 hours.

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TEST METHOD NO. 14

FORDING

1. TEST COURSE:

Body of fresh water not less than 36 inches deep and of sufficient size to permit the ditcher to enter.

2. TEST APPARATUS:

Equipment for collecting oil samples and analyzing samples for water content.

3. TEST PROCEDURE:

(a) Prior to the test, take oil samples from the engine and all hydraulic and lubricant reservoirs having parts less than 42 inches above ground level.

(b) Run the ditcher engine until the engine coolant reaches operating temperature as indicated by the temperature gage before the ditcher enters the water. The ditcher shall then be operated in first gear forward and reverse a minimum of 15 minutes in water at a depth of 36 inches as indicated by a line marked on the ditcher. Then operate the ditcher on the shore in high gear for 15 minutes.

(c) Perform the above test five times except that during the second, third, fourth, and fifth tests, the engine shall be started and stopped a minimum of two times while in the water at a 36-inch depth.

(d) On completion of the water fording tests, park the ditcher on a dry surface, shut off the engine, and take oil samples from the engine and all hydraulic and lubricant reservoirs for analysis as to water content.

(e) Analyze all samples to determine change in water content.

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TEST METHOD NO. 15

CROSS-COUNTRY-TYPE TRAVEL

1. TEST COURSE:

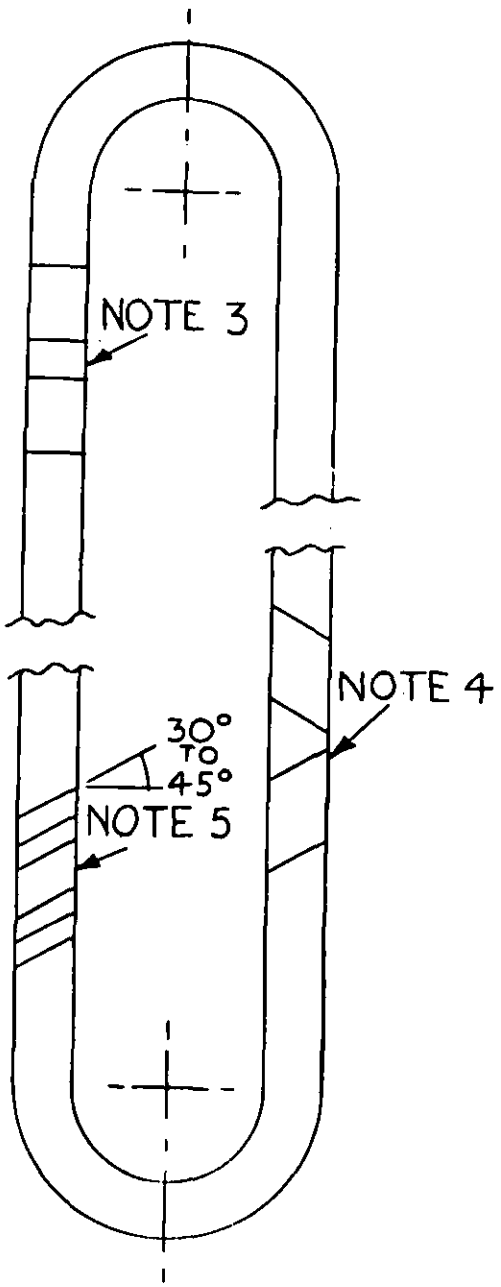
A figure eight or loop-shaped course shall be arranged over cross-country terrain which shall include 27 percent longitudinal grades and side slopes with a 17.5 percent grade. The length of the course shall permit a full lap in not more than 20 minutes. The course shall include a series of not less than 10 equally-spaced V-ditches cut at an angle to and across the line of forward travel. The ditches shall be arranged so that the right rear wheel and the left front wheel shall be in different ditches at the same time. The ditches shall be of sufficient depth to cause full oscillation of the front axle. The foregoing features shall be included in a course constructed generally in accordance with figure A-2.

2. TEST PROCEDURE:

Drive the ditcher over the course for a total of 18 hours at maximum speeds compatible with safety. Drive approximately one-half of the total time in each direction around the course.

NOTE: Crossties or humps may be installed in place of the ditches described above.

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NOTES:
1. COURSE MAY BE FIGURE 8 OR IRREGULARLY SHAPED TO SUIT TERRAIN.
2. LONGITUDINAL AND SIDE SLOPES MAY BE SEPARATE OR COMBINED.
3. 27% LONGITUDINAL GRADE, AT LEAST 100 FEET LONG ASCENDING AND 100 FEET LONG DESCENDING.
4. 17.5% SIDE SLOPES, LEFT AND RIGHT, AT LEAST 100 FEET LONG EACH.
5. NOT LESS THAN 10 V-DITCHES AS DESCRIBED IN TEST METHOD NO. 15.
6. LENGTH OF COURSE TO PERMIT A FULL LAP IN NOT MORE THAN 20 MINUTES.

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FIGURE A-2-CROSS-COUNTRY-TYPE TRAVEL COURSE

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TEST METHOD NO. 16

SIDEHILL SLOPES

1. TEST COURSE:

Terrain of undisturbed soil having a slope of not less than a 17.5 percent grade. The area shall be large enough to permit digging a series of ditches not less than 50 feet long for the required time.

2. TEST EQUIPMENT:

(a) Stopwatch for measuring digging time.

(b) Marked rod or pole to check digging depth.

3. TEST PROCEDURE:

Dig ditches for not less than 2 hours of sidehill operation across the slope described in 1 above. Do not include turnaround time as digging time. Dig ditches not less than 4 feet deep for approximately 45 minutes, not less than 5 feet deep for approximately 45 minutes, and 6 feet deep for approximately 30 minutes. Dig for approximately 1 hour with the right side of the ditcher facing up the slope, and approximately 1 hour with the right side facing down the slope. Operate the spoil conveyor to discharge to the uphill side of the ditch for approximately one-half of the digging time.

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TEST METHOD NO. 17

UPHILL SLOPES

1. TEST COURSE:

Terrain of undisturbed soil having a slope of not less than a 21 percent grade. The area shall be large enough to permit digging a series of ditches not less than 50 feet long for the required time.

2. TEST EQUIPMENT:

(a) Stopwatch.

(b) Marked rod or pole to check digging depth.

3. TEST PROCEDURE:

Dig for not less than 2 hours on uphill slopes at a 21 percent grade and at depths of from 4 to 6 feet. Do not include turnaround time as digging time. Dig ditches not less than 4 feet deep for approximately 45 minutes, not less than 5 feet deep for approximately 45 minutes, and 6 feet deep for approximately 30 minutes.. Operate the spoil conveyor to discharge approximately one-half of the time toward each side.

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TEST METHOD NO. 18

DOWNHILL SLOPES

1. TEST COURSE:

Terrain of undisturbed soil having a slope of not less than a 27 percent grade. The area shall be large enough to permit digging a series of ditches not less than 50 feet long for the required time.

2. TEST EQUIPMENT:

(a) Stopwatch.

(b) Marked rod or pole to check digging p depth.

3. TEST PROCEDURE:

Dig for not less than 2 hours on downhill slopes at a 27 percent grade and at depths of from 4 to 6 feet. Do not include turnaround time as digging time. Dig ditches not less than 4 feet deep for approximately 45 minutes, not less than 5 feet deep for approximately 45 minutes, and 6 feet deep for approximately 30 minutes. Operate the spoil conveyor to discharge approximately one-half the time toward each side.

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TEST METHOD NO. 19

CURVED TRENCH

1. TEST COURSE:

Level terrain of undisturbed soil. The area shall be large enough to contain the curved trenches described below.

2. TEST EQUIPMENT:

(a) Marked rod or pole to check digging depth.

(b) 100-foot tape.

3. TEST PROCEDURE:

At a 4-foot depth and without using the followup scraper, excavate a curved trench 270 degrees to the right and 270 degrees to the left at a radius of not more than 40 feet. Measure and record the radius of each curve to the centerline of the trench.

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TEST METHOD NO. 20

FOXHOLES

1. TEST COURSE:

Level terrain of undisturbed soil.

2. TEST EQUIPMENT:

Marked rod or pole to check depth of foxhole.

3. TEST PROCEDURE:

(a) Raise the followup scraper so that it does not operate or interfere with digging during this test. With the ditcher stationary, and starting with the digging boom in digging position and raised clear of the ground, operate the bucket-line at a speed suitable to the type of soil to be excavated. Operate the spoil conveyor. Dig a foxhole by lowering the digging boom vertically to maximum digging depth, then raising it clear of the ground. Move the ditcher 8 to 12 feet forward by the crowd drive. Check depth of foxhole.

(b) Excavate a series of not less than 11 more foxholes, following the procedure in 3(a) above.

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TEST METHOD NO. 21

DITCHES

1. TEST COURSE:

Level to gently rolling terrain of undisturbed soil of not less than three different soil types. The terrain shall contain an area of CL- or ML-type soil as defined by the Unified Soil Classification System (see 3.1.1), and not less than two additional types of soil as specified in 3.5.1.

2. TEST EQUIPMENT:

(a) Time recording device.

(b) 100-foot tape, stakes, hammer.

(c) Marked rod or pole to check digging depth.

(d) Complete operational log to record: Time, distances, digging depth, soil type, average rates of digging, maintenance performed.

3. TEST PROCEDURE:

Stake out a series of lines to guide the operator through the digging operation described below. Check digging depth not less than 4 times per hour during the test. Dig ditches not less than 100 feet long for a total of 200 hours and at depths ranging from 2 to 6 feet. At least 100 hours of digging shall be done at maximum depth and at least 65 hours at depths not less than 4 feet. During this test, dig not less than 12 ditches 100 feet long and 4 feet deep under conditions specified in 3.1.1. Measure and record the rate of ditching, in feet per minute, at the 4-foot depth. Divide the 100 hours digging at maximum depth evenly among the different soil types specified in 3.5.1.

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TEST METHOD NO. 22

RELIEF VALVE(S) SETTING

1. TEST EQUIPMENT:

   (a) Hydraulic pressure gage.

   (b) Timer.

   (c) Source of hydraulic oil under pressure, with pressure control (oil conforming to MIL-L-2104, grade 10).

   (d) Tools, couplings, and fittings needed to follow the test procedure below.

2. TEST PROCEDURE:

   (a) Connect the pressure gage and oil source into the circuit adjacent to the relief valve. Apply oil pressure to the relief valve in the pressure relief range, increasing the pressure at a rate of not more than 1 psi per second. Determine and record the pressure at which the valve "cracks" open and the pressure at full flow. Reduce oil pressure through the relief valve at a rate of not more than 1 psi per second. Determine and record the pressure when the valve reseats.

   (b) When more than one relief valve is provided, perform 2(a) above on all additional valves.

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TEST METHOD NO. 23

EASE OF MAINTENANCE

1. TEST FACILITY:

The test facility shall consist of an improved or paved surface or shop area of sufficient size to permit unobstructed surveillance of maintenance.

2. TEST APPARATUS:

(a) Common hand tools.

(b) Special tools (if furnished by the supplier with the ditchers).

3. TEST PROCEDURE:

(a) Examine the ditcher for accessibility of major assemblies.

(b) Itemize connectors, plugs and valves as to type and number of sizes.

(c) Examine position of drains with respect to accessibility.

(d) Activate drainage controls for coolant and lubricants. Observe path of discharge.

(e) Perform and time each operation on the maintenance operations list (see 3.7).

(f) Service the ditcher, noting and recording tools, equipment, and containers used.

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TEST METHOD NO. 24

EMERGENCY STEERING

1. TEST COURSE:

The test course shall be a dry, level surface of sufficient length for the ditcher to attain a speed of not less than 25 mph. The course shall also include a curve with a maximum centerline radius of 100 feet in accordance with figure A-3.

2. TEST APPARATUS:

(a) Tape, measuring (100-foot).

(b) Markers (20 each of sufficient length to extend 36 inches above ground level), equally spaced along curve centerline.

(c) Speed indicator.

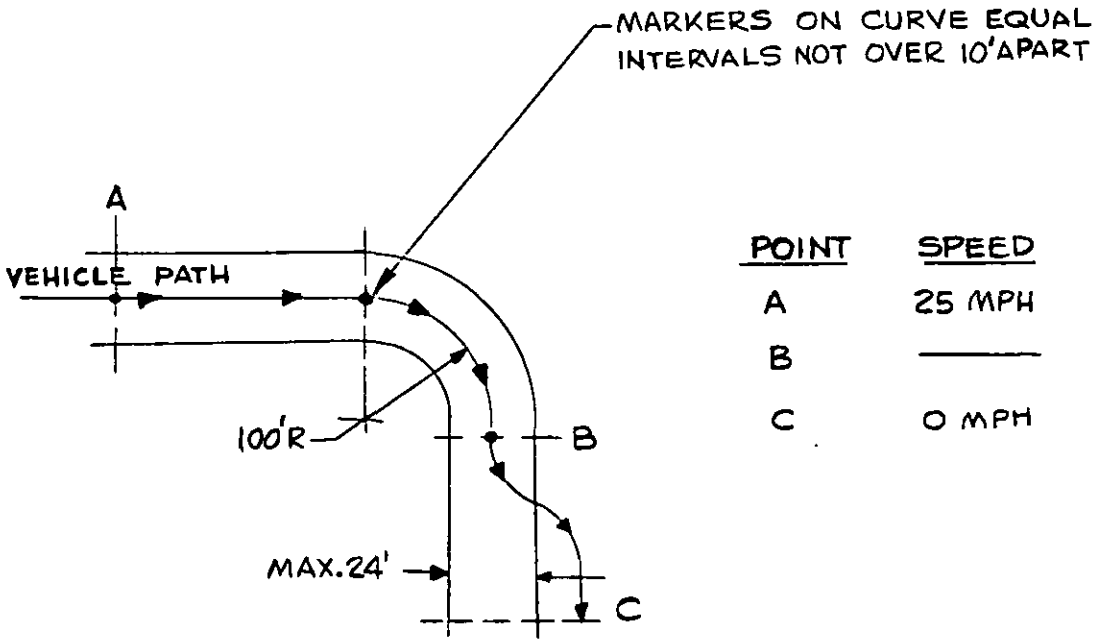
(d) Towing vehicle and tow cable.

3. TEST PROCEDURE:

(a) Drive the ditcher a sufficient distance to attain a speed of not less than 25 mph. When the ditcher reaches point A (figure A-3) traveling 25 mph, shut off the engine, shift the transmission into neutral. Slow the ditcher by use of the service brakes if necessary. At point B, steer the ditcher to the outside of the road away from the curve and pull off on the shoulder of the road. Stop at point C. The driver shall maintain complete control of the ditcher at all times during the test. All markers on the 100-foot radius curve shall be contacted by the ditcher while negotiating the curve.

(b) Attach the towing vehicle to the ditcher using the tow cable through the towing lugs on the front bumper of the ditcher. Do not start the ditcher engine; place the transmission in neutral. Tow the ditcher through the course (figure A-3) at a speed of 15 mph, steering as in (a) above. Stop at point C by using the ditcher brake system.

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FIGURE A-3. TEST COURSE FOR EMERGENCY STEERING TEST.

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TEST METHOD NO. 25

RAIN

1. TEST APPARATUS:

   (a) Timing device.

   (b) Metering device.

   (c) Spray fixture.

   (d) Wind generator and velocity gage.

2. TEST PROCEDURE:

   (a) Park the ditcher on a horizontal surface and expose to natural or simulated rain falling toward the ditcher at an angle of approximately 45 degrees to the horizontal. The volume of rain impinging on the horizontal surface shall be not less than 1/4 gallon per minute per square foot of wetted, horizontal area. Simulated rain shall originate from a source which produces droplets of uniform distribution and velocity similar to natural rain at the point of contact with the ditcher. The rain shall be driven onto and against the ditcher by a generated wind of not less than 30 miles per hour velocity. Conduct the test as follows:

      (1) Expose the top and left side to the rain as described above for a period of 5 minutes with the engine shut down.

      (2) With water flowing, start engine within 2 minutes (without removing accumulated water) and allow engine to idle for 2 minutes.

      (3) Repeat (1) and (2) with the top and front exposed.

      (4) Repeat (1) and (2) with the top and right side exposed.

      (5) Repeat (1) and (2) with the top and rear exposed.

   (b) At the conclusion of the test, examine the ditcher for the following:

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(1) Amount of water in the air-intake bowl and filters.

(2) Water leakage into the torque-converter oil chambers, engine crankcase sump, transmission, fuel tank, and hydraulic reservoir.

(3) Performance of the electrical components.

(4) The rate at which water accumulated in the exhaust system.

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TEST METHOD NO. 26

STORAGE

1. TEST FACILITY:

The test facility shall be a test chamber of sufficient size and capacity to contain the ditcher and to provide temperatures ranging from minus 65° to plus 155° F.

2. TEST APPARATUS:

(a) Thermocouples.

(b) Recording potentiometer(s).

3. TEST PROCEDURE:

(a) Low temperature storage.

(1) Subject the ditcher to a low ambient temperature until the temperature of all parts is stabilized at not higher than minus 65° F. Maintain the temperature of all parts continuously at minus 65° F. for a period of 12 hours.

(2) Examine the ditcher.

(b) High temperature storage.

(1) Subject the ditcher to a high ambient temperature until the temperature of all parts is stabilized at not lower than plus 155° F. Maintain the temperature of all parts at plus 155° F. for a period of 4 hours.

(2) Examine the ditcher.

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TEST METHOD NO. 27

FUNCTIONAL

1. TEST EQUIPMENT:

(a) Complete ditcher serviced and ready for operation.

(b) Dock or pit providing not less than a 6-foot clearance below the surface under the tires.

(c) Ramp or slope having an incline of not less than 20 degrees. The surface shall be either paved with concrete or asphalt or shall be compacted to a degree that will prevent the ditcher from making visible tracks when traveling up and down the slope.

(d) Heavy chain to restrain bucket-line.

2. TEST COURSE:

Level road or track with improved or paved surface.

3. TEST PROCEDURE:

Start the ditcher and perform the following operations to assure function of all components and to allow for minor adjustments:

(a) Back the ditcher to the edge of the dock or pit.

(b) Lower the digging boom assembly to the maximum digging depth.

(c) Operate the bucket-line in the digging direction, with engine at governed speed. Operate the bucket-line for 5 minutes in each of not less than 3 speeds, including the highest speed.

(d) Operate the conveyor belt throughout its entire speed range for periods of 5 minutes in each direction.

(e) Shift the spoil conveyor throughout its entire shifting range not less than 10 times.

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(f) Raise the digging boom assembly to travel position, then lower the boom assembly to maximum digging depth and raise it to travel position four more times.

(g) On the road or track, with the ditcher in travel position, place the controls in crowd drive and operate forward and backward. Starting from zero speed, increase the speed gradually to maximum speed in 1 minute, maintain maximum speed for 1 minute, then gradually decrease the speed to zero in 1 minute. Repeat the above operation in reverse travel direction. Perform the above crowd operation cycle three times.

(h) Drive the ditcher in travel position for not less than 2 miles over the road or track, shifting progressively from reverse to low to highest range and from highest to low to reverse not less than two cycles. During this driving operation, bring the ditcher to a full stop, from the maximum speed in high gear, with the service brakes not less than three times.

(i) Drive the ditcher forward up the ramp or slope. Stop the ditcher on the slope. Set the parking brake. Release the service brake, and place all controls in neutral, holding the ditcher on the slope with the parking brake only.

(j) Release the parking brake and drive the ditcher backward up the ramp or slope. Repeat the procedure in (i) above, except with the ditcher facing down the slope. Record the results of the above operations on a functional inspection check list prepared for each ditcher as follows.

(k) Repeat the procedure described in test method No. 1.

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FUNCTIONAL INSPECTION CHECK LIST

GENERAL DATA

Ditcher Make_____ Model No._____ Serial No._____

Engine Make_____ Model No._____ Serial No._____

___✓___ denotes satisfactory function

Digging Boom Assembly

_____Raise and lower boom assy., travel position to max. depth.

Bucket Line

_____Low speed

_____Intermediate speed

_____Highest speed

Conveyor Belt

_____Variable speed control

_____Reversing control

Conveyor Shift

_____Full range of travel

_____Shift direction reverse

Crowd Drive

_____Variable speed control

_____Direction reverse control

Road Travel

_____Gear shifting, reverse to highest range

_____Gear shifting, highest range to reverse

_____Service brake stop from maximum speed in high gear

Parking Brake

_____Facing up the slope

_____Facing down the slope

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TEST METHOD NO. 28

HYDRAULIC SYSTEM TEST

1. TEST APPARATUS:

(a) Pressure gages.

(b) Temperature gages.

(c) Timing device.

(d) Supply of filters and filter elements.

(e) Adjustable relief valve or hydraulic power source capable of developing a pressure equal to twice the system working pressure.

2. TEST PROCEDURE:

(a) With each line connected to its related component, insert filters in the circuits. For motors, connect the filter on the outlet side. Operate each function of the system for 5 minutes continuously, through full stroke for pistons, and at maximum rates. Visually examine all filters for contamination.

(b) If there is any evidence of contamination, repeat 2(a) until there is no evidence of contamination. If evidence of repeated contamination exists, disassemble that part of the system shown by the filters to be contaminated, clean, and examine part by part. Reassemble and repeat the procedure specified in 2(a) and 2(b). Reusable filters may be used for all but the last contamination check.

(c) Maintain the system oil temperature between 140° F. and 160° F., operate each control lever, and pressurize each pressure line for 15 seconds. Record each relief pressure and examine for external leakage.

(d) Connect each disconnected line to its related actuator. Disconnect the relief valve and connect the adjustable relief valve or hydraulic power source.

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(e) Pressurize each circuit to twice the working pressure and examine for leaks, deformation, or damage. Leakage due to defective seals and loose bolts may be corrected by replacement of seals and tightening of bolts to torque recommended by the supplier, providing no machining or welding is required. After repair of leaks has been made, the leakage test shall be repeated.

NOTE: Recording of relief pressures may be done at any time during the test.

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TEST METHOD NO. 29

TRAVEL SPEED

1. TEST COURSE:

The test course shall be a dry, level surface of sufficient length to attain maximum speed prior to entering measured distance.

2. TEST APPARATUS:

(a) Gage, tire pressure.

(b) Tape, measuring, surveyors.

(c) Watch, stop.

3. TEST PROCEDURE:

(a) Maintain the tires at pressure recommended by the supplier.

(b) Operate the ditcher engine at full throttle.

(c) Drive the ditcher a sufficient distance to attain maximum speed prior to entering the measured test course.

(d) Conduct a total of six runs, three in each direction in each gear as follows:

(1) 1st gear forward.

(2) High gear forward.

(3) Reverse gear.

(e) Travel speed in each gear shall be the average of six runs.

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TEST METHOD NO. 30

SOUND LEVEL

1. TEST FACILITY:

The test facility shall consist of an improved or paved surface or shop area of sufficient size to permit placement and surveillance of required instrumentation.

2. TEST APPARATUS:

In accordance with SAE J919.

3. TEST PROCEDURE:

In accordance with SAE J919.

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TEST METHOD NO. 31

HYDRAULIC COMPONENTS

1. TEST APPARATUS:

(a) Test bench with the following:

(1) Relief valve.

(2) Control valve - same as used on ditcher.

(3) Temperature gages and recorders.

(4) Torquemeter.

(5) Rpm meter.

(6) Flowmeter.

(7) Timer.

(8) Cycle counter.

(9) Hourmeter.

(10) Oil cooler.

(11) Dynamometer.

(12) Additional items to complete the system assembly.

(13) Power supply to produce oil pressures required.

(b) Oil conforming to MIL-L-2104, grade 10.

2. TEST PROCEDURE:

2.1 Hydraulic motor.

(a) Install the motor in the test system.

(b) Measure and record the overall efficiency (mechanical output/ hydraulic input) at the maximum speed in one direction as

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installed in the ditcher at an oil temperature of 150° F. plus or minus 5° F. and at a maximum of six pressure points at equal increments starting with 750 and continuing through 2,000 psi. Repeat the above for rotation in the opposite direction.

(c) Lock the motor shaft and with one port open to reservoir, pressurize the other port to 3,000 psi for 30 seconds. Check for leakage at the assembled surfaces and the shaft seals. Repeat, pressurizing the opposite port.

(d) Test the motor for torque output in both directions at maximum rated rpm with the pressure limited to not less than 2,000 psi.

(e) With the hydraulic motor in the test stand and the maximum rated load applied to the motor shaft, increase the oil pressure to the motor at a maximum rate of 1 psi per second. Note and record the oil pressure at the instant at which the motor shaft overcomes the rated load. Repeat the above for rotation in the opposite direction.

(f) Install the hydraulic motor in the system for running the endurance test cycle as follows: 30 minutes clockwise and 30 minutes counterclockwise direction for a total of 500 hours. Motor shall be endurance tested at maximum installed rpm, maximum rated load, and with the pressure limited to not less than 2,000 psi. For the first 50 hours, the oil temperature shall be 220° F. plus or minus 5° F. The oil temperature at the motor inlet shall be 150° F. plus or minus 5° F. for the remaining 450 hours of the test.

(g) Repeat 2.1(b), 2.1(c), 2.1(d), and 2.1(e) above.

(h) After completing the above tests, disassemble and examine the motor.

2.2 Hydraulic suction line.

(a) Soak the reservoir-to-pump suction line assembly in oil for 5 hours at 190° F. plus or minus 10° F.

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(b) Remove the line assembly from the oil bath, attach to the pressure connection of the test bench, and blank off one end. Measure and record the length of the hose. Pressurize the line assembly to 50 psi, hold this pressure for 15 seconds; then measure and record the length of the hose and examine for leakage.

(c) Draw a vacuum of 7 in Hg. and hold for 30 seconds. Examine for leakage.

(d) Subject the hose to 100 psi.

(e) Disassemble the line assembly and examine for defects and deterioration.

2.3 Hydraulic pump.

Procedure per MIL-P-52675.

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APPENDIX, PART II

DATA QUESTIONNAIRE

Questionnaire for Physical and Functional Data of Ditching Machine

PART I - DESCRIPTION

The Ditcher submitted is described as follows:

Supplier -

Supplier's Model Designation -

Serial Number Range - Thru

A. ENGINE

Make -

Model -

Series -

Type -

Cycles -

No. Cylinders -

QPL RATED VALUES

Maximum Flywheel Horsepower: at Governed RPM

Intermittent " " : at Governed RPM

Continuous " " : at Governed RPM

AS INSTALLED

Maximum Flywheel Horsepower: at Governed RPM

Intermittent " " : at Governed RPM

Continuous " " : at Governed RPM

Bore -

Stroke -

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Injector Size -

Compression Ratio -

Air Induction System -

Suppliers shall furnish certified torque and net horsepower output curves for intermittent duty. Suppliers shall furnish certified converter performance curves for engine converter output.

B. ENGINE ACCESSORIES

1. Air Cleaner

a. Make -

b. Model -

c. Type -

2. Oil Filter

a. Type -

b. Make -

c. Housing No. -

d. Element No. -

3. Fuel Filter

Primary:

a. Type -

b. Make -

c. Element No. - : FSN

Secondary:

a. Type -

b. Make -

c. Element No. - : FSN

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4. Temperature Gage

   a. Type -

   b. Make -

   c. Range Degrees F. to Degrees F.

   d. Location -

5. Fuel Pump

   a. Type -

   b. Make -

   c. Capacity -

   d. Element No. : FSN

6. Hourmeter

   a. Type -

   b. Actuated by -

   c. Make -

   d. Model -

7. Governor

   a. Unit

      (1) Type -

      (2) Make -

      (3) Model -

      (4) Setting for engine rpm.

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8. Muffler (when applicable)

   a. Type -

   b. Make -

9. Rain Protection for Exhaust -

10. Starting System, Electric

   a. Type -

   b. Voltage -

   c. Make -

   d. Model -

11. Lubricating Oil Pressure Gage

   a. Type -

   b. Range to psi.

12. Hand Throttle (if applicable)

   a. Type -

   b. Location -

13. Turbocharger (if applicable) - N.A.

   a. Type -

   b. Make and Model -

   c. Boost Ratio

   d. Maximum recommended turbine speed rpm.

C. COOLING SYSTEM

1. General

   a. Type -

   b. Overall capacity gal.

   c. Pressurization psi.

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2. Radiator

   a. Type -

   b. Capacity gal.

   c. Make -

   d. Serial No. -

   e. Protective guard -

3. Fan

   a. Type -

   b. Diameter in.

   c. No. Blades -

   d. Blade spacing -

   e. Horsepower rating -

4. Water pump

   a. Type -

   b. Make and Model -

   c. Drive -

   d. Rating - gpm at rpm.

   e. Speed as installed rpm; gpm.

5. Thermostat

   a. Type -

   b. Make and Model -

   c. Temperature rating: Full open - degrees F.
   Closed - degrees F.

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D. POWER TRAIN

1. Transmission

a. Type -

b. Make -

c. Speed range: Forward: Reverse:

d. Gear ratios:

Forward: 1 2 3 4 5 6

Reverse: 1 2

e. Filter -

(1) Type -

(2) Make -

(3) Housing No. -

f. Element No.: FSN

2. Transfer Case

a. Type -

b. Make -

c. Model -

d. Range -

e. Speed ranges: Forward: Reverse:

3. Torque Converter

a. Make -

b. Model -

c. Torque input - rated: actual:

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d. RPM -

e. Horsepower Input - rated: actual:

f. Torque multiplication factor -

g. Heat exchanger -

4. Axles

a. Front

(1) Type -

(2) Make -

(3) Model -

(4) Capacity -

(5) Serial No. -

(6) Mounting -

b. Rear

(1) Type -

(2) Make -

(3) Model -

(4) Capacity -

(5) Serial No. -

(6) Mounting -

E. FINAL DRIVE

1. No. driving wheels: front_____rear_____

2. Type drive -

3. Final drive gear ratio -

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4. Steering gear -

a. Make -

b. Model -

c. Type -

F. BRAKES

1. Service Brake - Primary System

a. Type -

b. Make -

c. Model -

d. Diameter of brake drums_____No._____

e. Lining width -

f. Brake chamber volume cu. in.

g. Location - No. of wheels affected

h. Air compressor: Type Capacity

Type Drive Location

Make Model

i. Reservoir size cu. in.

j. Operating pressure psi

k. Pressure gage Location

2. Parking Brake

a. Type -

b. Make -

c. Model -

d. Actuating - Wheels

e. Control: Type Located

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G. CONTROLS

1. Operating

a. Type -

b. Overload protection -

2. Steering Gear

a. Type -

b. Make -

c. Model -

d. Location -

3. Hydraulic System

a. Make -

b. Tank capacity gal.

c. Filler location -

d. Oil tank breather location -

4. High Pressure Lines

a. Type -

b. Protection -

5. Pressure Relief Valve (1)

a. Type -

b. Make -

c. Piston size -

d. Pressure setting -

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6. Pump(s)

a. Type -

b. Make -

c. Model -

d. Drive -

e. Drive ratio, engine to pump

f. Capacity - Rated: Actual:

(1) GPM

(2) At RPM

(3) PSI

7. Filter

a. Make -

b. Model -

c. Housing No. -

d. Element No. - :FSN

H. OPERATOR'S PLATFORM AND SEAT

1. Platform

Material

2. Seat (operator's)

a. Type -

b. Dimensions -

c. Material -

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3. Backrest (operator's)

a. Type -

b. Dimensions -

4. Seat (driver's)

a. Type -

b. Dimensions -

c. Material -

5. Backrest (driver's)

a. Type

b. Dimensions

I. TIRES

Type -

Make -

Size -

Ply rating -

PSI -

Tread -

J. ELECTRICAL EQUIPMENT

1. Battery

a. Number required -

b. Size -

c. Volts each -

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d. Make -

e. Rating - amp hr at hr rate

f. No. Plates -

g. Mounting - Location

2. Generator or alternator (designate which)

a. Type -

b. Volts -

c. Rating watts at rpm.

d. Make -

e. Model -

f. Drive -

g. Control: Type -

Make -

Model -

h. Reverse polarity protection (if alternator)

i. Serial No. -

3. Horn

a. Type -

b. Make -

c. Button location -

4. Starter

a. Type -

b. Make -

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c. Model -

d. Serial No. -

5. Electric wiring

a. Type -

b. Protection -

K. GAGES AND INSTRUMENTS

1. Engine oil pressure

a. Make -

b. Model -

c. Range -

d. Location -

2. Engine temperature gage

a. Make -

b. Model -

c. Range -

d. Location -

3. Engine hourmeter -

a. Make -

b. Model -

c. Range -

d. Location -

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4. Ammeter -

   a. Make -

   b. Model -

   c. Range -

   d. Location -

5. Torque converter temperature gage (if applicable)

   a. Make -

   b. Model -

   c. Range -

   d. Location -

6. Torque converter pressure gage (if applicable)

   a. Make -

   b. Model -

   c. Range -

   d. Location -

7. Air pressure gage

   a. Make -

   b. Range -

   c. Location -

L. TOOLBOX(ES)

Type -

Number -

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Dimension -

Location(s) -

Cover Fastening -

M. FUEL TANK

Capacity gals

Location -

Drain plug location -

PART II. PHYSICAL AND FUNCTIONAL DIMENSIONS

A. PHYSICAL DIMENSIONS

1. Shipping weight and standard equipment, Dry lb.

2. Operating weights with fuel, coolant, oil, grease, hydraulic fuel, etc. (w/operator):

   a. Front lb.

   b. Rear lb.

   c. Total lb.

3. Wheelbase inches

   a. Tread inches

4. Towing lug pad eyes

   a. Capacity lb.

   b. Height above ground inches

5. Turning clearance diameter (turning clearance diameter is defined as the diameter of the smallest circle described by the outermost part of the ditcher in making a 180-degree turn).

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a. Right turn feet.

b. Left turn feet.

6. a. Overall length, ditching mechanism on level ground inches.

b. Overall length, ditching mechanism in traveling position inches.

c. Overall width inches.

d. Width between centerline of tires inches.

e. Overall height, ditching mechanism on level ground inches.

f. Overall height, ditching mechanism in traveling position inches.

g. Overall height, minimum reducible inches.

h. Ground clearance, under chassis inches.

Ground clearance, under boom with bucket teeth at lowest position inches.

i. Center of gravity location, vertical inches

longitudinal inches.

7. Buckets

a. Number -

b. No. of teeth on each bucket -

c. Method of installing teeth -

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8. Digging operation

a. Maximum depth -

b. Digging rate at 4 ft. depth feet per minute.

c. Minimum digging diameter at 4 ft. depth feet.

d. Conveyor speed, minimum feet per minute.

e. Conveyor speed, maximum feet per minute.

PART III. Supplier shall furnish detailed list of all items which vary from their normal commercial ditcher.

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U. A. ARMY MOBILITY EQUIPMENT COMMAND
RESEARCH, DEVELOPMENT AND ENGINEERING DIRECTORATE
ATTN: AMSME-RZK-KX
FORT BELVOIR, VIRGINIA 22060

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-D-52091C(ME) - Ditching Machine: Diesel-Engine-Driven; Wheel-Mounted; Ladder

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

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